

# ADVANCED VOID AUTOMATION

## The 3D Factory Field Guide

Build 0.2.41 | Minecraft Bedrock

BUILD / AUTOMATE / DISCOVER | Updated block crafting recipes and material conversions

From your first sapling to MV automation, seared casting, advanced factories and Supremium armor.

Recipe data exported from the actual pack. Build scenes rendered from pack geometry and textures, with official Mojang terrain artwork.

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# Quick start for your first island

Import the .mcaddon into Minecraft Bedrock. Activate the matching Behavior and Resource Packs in your skyblock world and enable Beta APIs. Back up an existing world before updating it.

This walkthrough starts on an island with a tree and a patch of dirt. “Use” means right-click, tap Interact, or press your controller’s interact button. Save one sapling for the next tree and keep the original dirt to plant it on.

|  |  |  |  |
|---|---|---|---|
| *Wood Crook* | *Silkworm* | *Sieve*<br>Crafting recipe | *Pebble* |

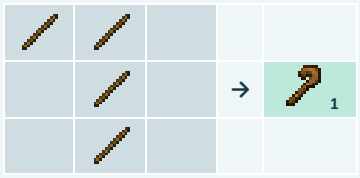
**1. Make tools from the first tree.** Turn Oak Logs into Oak Planks. Make a Crafting Table from four planks, then craft sticks from two planks stacked vertically. Make a Wood Crook before breaking the leaves.

**2. Grow more trees and get string.** Break normal oak leaves with the crook for saplings and a chance at Silkworms. Use a Silkworm on an intact oak leaf block, wait for nearby oak leaves to become infested, then crook the infested leaves for String.

**3. Build a Sieve and Composter.** Craft the Sieve with four Oak Planks and four String in any positions. It comes with a basic mesh. Make a regular Composter from seven wooden slabs in a U shape.

**4. Make your first Cobblestone.** Use eight spare saplings on the empty Composter. When the green filling turns to dirt, use it with an empty hand to collect one Dirt. Use the Dirt on the Sieve, then keep interacting until the batch finishes. The loading interaction counts as the first of six. Collect the pebbles and craft four ordinary Pebbles together for one Cobblestone. Repeat to build a stockpile.

**Wood Crook**



**Wood Hammer**

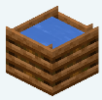


**Composter (wooden slabs)**



***GO TO*** *Sieve* | *Dirt build* | *Sieve drop rates* | *Next: make water*

# Make water, clay and your first bucket

|  |  |  |  |
|---|---|---|---|
| *Rainwater Composter* | *Dust* | *Clay Ball* | *Clay Bucket* |

**1. Collect rain for your first water.** Place an empty Composter under open sky in the Overworld. Rain fills it in four stages. Leave it on ordinary ground so the water stays fresh. A full Composter holds 1,000 mB, or one bucket.

**2. Make water any time with leaves.** Use shears to collect four ordinary leaf blocks and add them to an empty Composter, one at a time. Each adds 250 mB. To make shears, hammer Cobblestone into Gravel and sieve it for Iron Fragments. Craft four fragments into one Raw Iron and smelt it in a Furnace made from eight Cobblestone around an empty center. Fuel it with wood. Repeat to get the two ingots for shears.

**3. Make Clay.** Use a Wood Hammer on Cobblestone, then Gravel, then Sand to get Dust. Use one Dust on the filled Composter to turn the entire water fill into one Clay Block. Break the Clay Block to get four Clay Balls.

**4. Make a bucket and refill the Composter.** Shape three Clay Balls into an Unfired Clay Bucket and fire it in a regular Furnace. Refill the Composter with rain or leaf blocks, then use the empty Clay Bucket on it to draw the water. An Iron Bucket crafted with three Iron Ingots also works.

| **Unfired Clay Bucket** | **Make Mud** | **Make Grout** |
|---|---|---|
|  | Use one Dust on a full ordinary-water Composter. This uses all 1,000 mB. | Craft one Clay Ball, one Sand and one Gravel together to get two Grout. |

For a repeatable water supply later, place a Moisture Collector on solid ground and put an empty Composter directly above it. Feed the collector a Lava Bucket or connect a lava Tank. It uses 10 mB of lava to produce 50 mB of water per second while there is room for the water.

***GO TO*** *Water and clay illustration* | *Pumped water build* | *Next: make lava*

# Make lava and a renewable cobblestone supply

|  |  |  |  |
|---|---|---|---|
| *Kiln* | *Tank* | *Fluid Pump* | *Stone Barrel* |
| Crafting recipe | Crafting recipe | Crafting recipe | Crafting recipe |

**1. Craft a Kiln.** Place eight Cobblestone in the Crafting Table, leaving the bottom-center slot empty. It needs no power or external heat.

**2. Melt Cobblestone.** Hold extra Cobblestone and use it on the placed Kiln. Each piece yields 1,000 mB of lava, or one bucket, after about 10 seconds at normal tick speed.

**3. Take the lava.** Use an empty Clay Bucket or Iron Bucket on the Kiln once it has at least 1,000 mB. Keep lava in a nonflammable area away from your tree and wood platform.

**4. Make a cobblestone generator.** Dig a trough in a fireproof platform. Place a water source at one end and put a one-block-deep catch pit in the next cell. Leave an open meeting cell beyond the pit, then place the lava source at the far end. Water falls into the pit and flowing lava reaches the water from the other side. Mine the Cobblestone as it forms.

| If you need | Do this |
|---|---|
| Trough from above | Water source \| Deep catch pit \| Open mining cell \| Lava source. Build a floor and solid walls around the trough to contain the flow. |
| A bigger lava supply | Place a Fluid Pump by the Kiln and run purple Transfer Pipes to a compatible Tank. |
| A way to store water | Use a separate Tank group for water. Connected Tanks share storage, so keep the water and lava groups apart. |
| A Stone Barrel | Craft seven Cobblestone in a U. Fill it with lava for the recipes on the next page. |

***GO TO*** *Kiln* | *Stone Barrel* | *Pumps and tanks* | *Next: lava and Witch Water recipes*

# Make Obsidian, Netherrack, End Stone and Soul Sand

A Stone Barrel holds lava only. Use a Lava Bucket on an empty Stone Barrel. Each recipe uses 100 mB, so a full 1,000 mB bucket can supply ten conversions.

| Output | Use this ingredient on the lava-filled Stone Barrel |
|---|---|
| 1 Obsidian | 1 Water Bucket or Clay Water Bucket. The water is a recipe ingredient and you get the empty bucket back. |
| 1 Netherrack | 1 Redstone Dust. Sieve Sand or Dust to get Redstone. Hammer Netherrack into Crushed Netherrack for further sieving. |
| 1 End Stone | 1 Glowstone Dust. Sieve Sand or Dust to get Glowstone Dust. |

## Make Witch Water and Soul Sand

**1. Make Mycelium.** Sieve Sand or Red Sand for Ancient Spores. The base roll is 5% per input with the basic mesh. Use the spores on a Dirt block to turn it into Mycelium.

**2. Brew Witch Water.** Place a regular Composter directly on the Mycelium. Fill it with one bucket of water or four normal leaf blocks. Wait for the status to say Witch Water. Keep this setup separate from your ordinary water supply.

**3. Make Soul Sand.** Use one Sand on the finished Witch Water. Collect the Soul Sand and refill the Composter to brew another batch. You can sieve the Soul Sand for Quartz and a chance at Nether Wart or a Ghast Tear.

To start the smeltery, smelt Grout in a Furnace to get Seared Brick items, then craft four of them in a square to make one Seared Bricks block.

***GO TO*** *Stone Barrel illustration* | *Witch Water illustration* | *Build the smeltery* | *Contents*

# From hand tools to your first factory

## 1. Make enough metal for a power line

Start with the basic Sieve and a renewable Cobblestone supply. Hammer Cobblestone into Gravel for Coal, Iron Fragments, Nickel Fragments and Aluminum Scrap. Hammer Gravel into Sand for Crushed Copper. Sieve Dust or Sand for Redstone Dust.

Craft four Nickel Fragments into a Nickel Ingot and four Aluminum Scraps into an Aluminum Ingot. Craft four Crushed Copper into one Raw Copper, then smelt it in a Furnace. Make two Copper Ingots for your first Combustion Generator.

Craft and place the generator, add logs or charcoal to its fuel slots, and run red Watt Pipes from it to one machine. Start with a short line. The generator can supply its stored energy while the flame is idle.

***CRAFT*** *Combustion Generator* / *Watt Pipe*

## 2. Get one item route working

Put a test stack in a chest. Attach a Transfer Node with its intake facing that chest, run purple Transfer Pipes, and end at a second chest with free space. Once the stack moves, reuse the same source-to-destination layout to feed inputs or collect output from a machine.

***BUILD*** *Transfer Node* / *Transfer Pipe* / *Item routing and filters*

## 3. Build a one-layer smeltery and cast your first ingot

Start with 21 Seared Bricks blocks for the floor and nine for the walls. Leave the four outer corners open on both levels. Fill three wall positions with the Controller, Drain and Seared Tank. Leave the 3 × 3 center above the floor empty.

Place the Faucet on the outside of the Drain and the Casting Table directly below the Faucet. The 30 Seared Bricks blocks and five functional pieces take 152 individual Seared Bricks, plus one Furnace for the Controller and one Glass for the Tank. Smelt 152 Grout to make the brick items.

Fill the Seared Tank with lava. Craft a Stone Ingot Pattern from two Cobblestone side by side and place it in the Casting Table. Melt two Gold Ingots, select Gold on the Controller and use the Faucet. Collect the reusable Gold Ingot Cast and put it back in the Table. Then melt and select another metal. Pour one ingot’s worth and collect it once cooled.

***BUILD*** *Smeltery layout* / *Smeltery Controller* / *Smeltery Drain* / *Seared Tank* / *Seared Faucet* / *Casting Table*

## 4. Unlock MV Storage

Use the smeltery to make Steel. Make Netherite from sieved Crushed Netherrack, save nine Diamonds for a Diamond Block, and collect the other metals on the MV recipe card. Craft the Universal Crafter first, then use it as an ingredient for MV Storage. Power MV with Watt Pipes and test deposit and withdrawal before adding MV Links.

***CRAFT*** *Universal Crafter* / *MV System* / *Steel and Netherite routes* / *Connect MV Storage*

# 1. Start here: installation and using this handbook

| | | | |
|---|---|---|---|
|  <br>*MV System*<br>Crafting recipe |  <br>*Sieve*<br>Crafting recipe | | |

Advanced Void Automation 0.2.41. Illustrated survival and automation handbook with the latest power, smeltery, reactor and flying ship updates.

Import the mcaddon and activate BOTH Behavior and Resource Packs at the same version. Keep the Script API/Beta APIs required by the current pack enabled. Restart Minecraft if the library still displays an older cached version.

Existing worlds do not need to be deleted. Back up and close the world, then update both embedded packs or select both matching imported versions. An old embedded pack and a newer shared-library pack can appear separately in the version list.

For native Supremium flight, enable Education features and Cheats. The other factory systems do not require Education. A normal addon cannot switch on smooth native Creative flight through the read-only Player API alone; this build does not silently restore the old jittery simulation.

Use the clickable contents to jump to a system, the machine directory to identify a block, and the recipe index for the added-block crafting grids. Shaped grids show exact positions; shapeless grids show ingredients. Construction images are labeled schematics. Item and ingredient pictures use current pack textures.

The handheld Wireless MV Remote binds directly to MV Storage. Use the wired MV Link Unit to connect item automation to storage.

***JUMP TO:*** *Machine directory* | *Block recipes* | *Drop tables* | *Essence equipment recipes* | *Tree to Supremium costs*

## 2. First island: wood, saplings, string and cobblestone

| | | | |
|---|---|---|---|
|  |  |  |  |
| *Wood Crook* | *Silkworm* | *Sieve*<br>Crafting recipe | *Pebble* |

Protect the initial tree and saplings. Expand before harvesting everything so drops do not fall into the void. Make a crafting table, sticks and a Wood Crook. Replant before using the last sapling.

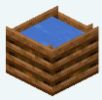
Use a crook on normal leaves for its species-matching bonus sapling roll and Silkworm roll. Shared bonus rolls are 15% sapling and 3% Silkworm per leaf, in addition to native leaf loot. Better crooks mainly improve speed/durability; they do not change those bonus odds.

Use a Silkworm on oak leaves to infest them. Infestation spreads to neighboring oak leaves. Harvest infested leaves for string and establish the sieve/mesh progression. Sneak vein mining applies the same rolls separately to every leaf.

Use eight surplus saplings in an empty Composter for one Dirt. Sieve Dirt for ordinary Pebbles, then craft four together into one Cobblestone. Specialty pebbles make their own stone types.

***JUMP TO:*** *Machine directory* | *Block recipes* | *Drop tables* | *Essence equipment recipes* | *Tree to Supremium costs*

# 3. Composting, water, clay and mycelium

| | | | |
|---|---|---|---|
|  |  |  |  |
| *Rainwater Composter* | *Moisture Collector*<br>Crafting recipe | *Ancient Spores* | *Mycelium* |

For Dirt, use eight saplings in an empty Composter. Wait for the green filling to finish, then use an empty hand to collect the Dirt.

For your first water, collect rain in an empty Composter under open sky or add four ordinary leaf blocks. Use an empty bucket on the full Composter, or use one Dust to make a Clay Block or one Dirt to make Mud.

For renewable water later, place a Moisture Collector on solid ground with a Composter directly above it. Supply lava to the collector. Use a Fluid Pump to move ordinary water into a dedicated water Tank.

For Witch Water, sieve Sand or Red Sand for Ancient Spores, use the spores on Dirt, then place a water-filled Composter directly on the Mycelium. Use Sand once it becomes Witch Water to make Soul Sand.

The Stone Barrel holds lava for Netherrack, End Stone and Obsidian. Use the regular Composter for water-based clay and mud recipes.

***STEP BY STEP*** *Make water and clay* | *Make lava* | *Barrel and Witch Water recipes*

***BUILD EXAMPLES:*** *Eight saplings into dirt* | *Witch Water into Soul Sand* | *Water conversions: clay and mud*

***JUMP TO:*** *Machine directory* | *Block recipes* | *Drop tables* | *Essence equipment recipes* | *Tree to Supremium costs*

# 4. Hammers, sieving and mesh progression

| | | | |
|---|---|---|---|
|  |  |  |  |
| *Wood Hammer* | *Dust* | *Iron Sieve Mesh* | *Netherite Sieve Mesh* |

Use a hammer for supported crushing conversions. The early chain starts Cobblestone -> Gravel -> Sand -> Dust. Other supported rocks have their own outputs. A better hammer does not automatically improve sieve odds.

Use a supported block on a Sieve, then keep interacting until its six work clicks complete. The loading interaction counts as the first click. Collect the physical drops on a safe platform.

Use Iron, Diamond or Netherite Mesh on the sieve to upgrade it. Basic/String, Iron, Diamond and Netherite apply x1, x1.25, x1.6 and x2 to independent base rolls, capped at 100%. Sneak-use empty-handed to remove an upgrade and return the mesh.

Dirt has a 5% base Inferium Essence roll. With Iron, Diamond and Netherite Mesh that becomes 6.25%, 8% and 10%. The drop tables include the weighted fallback used when every independent roll fails.

***BUILD EXAMPLES:*** *A compact sieve workshop*

***JUMP TO:*** *Machine directory* | *Block recipes* | *Drop tables* | *Essence equipment recipes* | *Tree to Supremium costs*

# 5. Fragments, crushed ore, powder and ingots

| | | | |
|---|---|---|---|
|  |  |  |  |
| *Iron Fragment* | *Crushed Iron Ore* | *Iron Ore Powder* | *Iron Ingot* |

Gravel yields fragments, Sand yields crushed ore, and Dust yields powders. For iron, copper and gold, combine four matching fragments or crushed pieces into one Raw Ore and smelt it in a Furnace. Four matching powders craft one ingot directly. Custom metals can use their own combine recipes.

Fresh raw ore processing can have an ore bonus, but recycled ingots must not multiply forever. Four Iron, Gold or Copper Powders combine into an INGOT, not raw ore. Pulverizing a finished ingot and recombining its powder is neutral recycling.

The Smeltery accounts for powder, ingots and fresh raw ore using different molten quantities. Four powders represent one ingot. Supply machine fuel/power and free output space before assuming a processor has stalled.

***JUMP TO:*** *Machine directory* | *Block recipes* | *Drop tables* | *Essence equipment recipes* | *Tree to Supremium costs*

# 6. Generators, Watt Pipes and battery backup

| | | | |
|---|---|---|---|
|  |  |  |  |
| *Combustion Generator* | *Solar Panel* | *Pink Slime Generator* | *Battery Bank* |
| Crafting recipe | Crafting recipe | Crafting recipe | Crafting recipe |

A Combustion Generator generates 80 W per working second. Its 10,000 W buffer continues supplying connected machines during standby and while remaining charge lasts after fuel runs out. It resumes burning once there is room for another 80 W. If a powered machine stops while the generator is charged, check its own status and Watt Pipe connections.

Red Watt Pipes form the electrical network. Purple Transfer Pipes carry items/fluids. Choose a generator supported by your supplies: Combustion, Solar/Advanced Solar, lava conversion or the late Pink Slime route. Read the live HUD for power and fuel conditions.

An empty source cannot power machines using stale pipe energy. Solar depends on its active lighting/day conditions. Check source fuel, red pipe connections and workload when a machine reports no power.

Place touching Battery blocks to combine their storage. Connect the bank to sources through Watt Pipes. Multiple sources can contribute spare charging power; full banks stop charging. Charge remains through reloads.

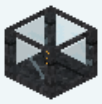
Batteries remain standby while primary generation is active, then discharge when primary sources go down. They do not supplement an undersized but still-active primary source. Remove source fuel for a safe outage test.

The connected bank displays a centered horizontal gauge on one front face, red when low and green when high, plus numeric status. A broken battery drops empty; plan storage changes before dismantling a charged bank.

***BUILD EXAMPLES:*** *A battery bank on a shared power line*

***JUMP TO:*** *Machine directory* | *Block recipes* | *Drop tables* | *Essence equipment recipes* | *Tree to Supremium costs*

# 7. Fluid Pumps, tanks and lava handling

| | | | |
|---|---|---|---|
|  |  |  |  |
| *Fluid Pump* | *Tank* | *Seared Tank* | *Kiln* |
| Crafting recipe | Crafting recipe | Crafting recipe | Crafting recipe |

For water, place an empty Composter on top of a lava-fed Moisture Collector. Use a Fluid Pump and purple Transfer Pipes to connect the source to a compatible water Tank. A full tank or a tank containing another fluid cannot accept more.

For lava, use Cobblestone on the Kiln. Each piece melts into one bucket of lava without power. Use an empty bucket on the Kiln or connect a Fluid Pump and a dedicated lava Tank.

The Kiln is an early lava route. Supply its supported solid input and connected output storage, then wait for processing. A blocked/full output tank can prevent further production.

Seared Tanks supply the smeltery fuel arrangement. Do not mix different fluids into an occupied group. Molten metals can ignite entities like lava; build safe walkways rather than stepping into the molten volume.

***BUILD EXAMPLES:*** *Water collector, pump and tank*

***JUMP TO:*** *Machine directory* | *Block recipes* | *Drop tables* | *Essence equipment recipes* | *Tree to Supremium costs*

# 8. Transfer Nodes, filters, priorities and bulk storage

| | | | |
|---|---|---|---|
|  |  |  |  |
| *Transfer Node* | *Transfer Pipe* | *Polymer Circuit Board* | *Black Hole Unit* |
| Crafting recipe | Crafting recipe | | Crafting recipe |

Place a Transfer Node against the source inventory and route purple Transfer Pipes to a destination. Keep intake direction unambiguous. Sneak-use rotates the intake. Start with one source chest and one destination when troubleshooting.

Use a Polymer Circuit Board on a node to enable smart routing. Open its HUD and place actual example items into persistent filter slots. Remove them manually when you want to change the filter.

ALLOW listed items transfers only matching items; BLOCK listed items excludes them. An empty filter permits all items. The button label names the active mode. Destination priority tries High, then Normal, then Overflow, skipping full or incompatible inventories.

A Black Hole Unit can accept and supply items through the transfer network. Its capacity does not remove the need for a correct source/destination route. A Vacuum Hopper collects loose drops; conveyors and Express Conveyors move physical items. Filter overflow after reserving production inputs.

***HUD EXAMPLES:*** *Smart Transfer Node: filters are not output slots*

***BUILD EXAMPLES:*** *Filtered chest-to-chest transport* | *Bulk storage off a farm line*

***JUMP TO:*** *Machine directory* | *Block recipes* | *Drop tables* | *Essence equipment recipes* | *Tree to Supremium costs*

# 9. MV Storage, in-screen crafting and wireless access

| | | | |
|---|---|---|---|
|  <br>*MV System*<br>Crafting recipe |  <br>*MV Link*<br>Crafting recipe |  <br>*Wireless MV Controller* | |

Build and power an MV System. Open its custom storage HUD, search stored items with Find, and use its deposit/withdrawal controls. Selected withdrawal amount is separate from the total stored quantity.

The 3x3 manual crafting pattern is inside MV Storage. Place a pattern and use its output to craft from stored ingredients. The removed recipe-search controls and separate MV Pattern block are not required.

MV Links connect item networks to storage. Choose Transfer or Pull as needed. Filters, pauses and reach still apply. A linked automatic crafter may need both input and output directions permitted.

Sneak-use the Wireless MV Remote on the desired MV to bind it, then use the remote normally to open that MV HUD. MV, storage and power must be loaded in the same dimension. The remote does not force-load chunks. The retired Wireless MV Link access point is not required. Keep the wired MV Link Unit for automated item transfers.

Empty valuable storage and internal inventories before dismantling. Do not delete hidden inventory hosts to force a machine to break; they can hold real items.

***HUD EXAMPLES:*** *MV Storage: storage totals, recipe pattern and player inventory*

***BUILD EXAMPLES:*** *MV Storage with a handheld remote*

***JUMP TO:*** *Machine directory* | *Block recipes* | *Drop tables* | *Essence equipment recipes* | *Tree to Supremium costs*

# 10. Universal Crafter and MV automation

| | | | |
|---|---|---|---|
|  *Universal Crafter* Crafting recipe |  *Crafting Planner* | | |

The Universal Crafter uses a 3 x 3 sample recipe. Drag ingredients into the grid, then return the remaining cursor stack to your inventory before using its controls. Samples are returned; recipes consume ingredients from the connected MV storage.

Click Add automation to save and start a valid recipe immediately. No preliminary manual craft is needed. The sample grid clears after adding. Up to ten saved recipe boxes appear in Recipe / Queue. Click a box to stop, resume, remove or load it.

Mode: Auto crafts continuously while ingredients, MV room and power are available. Optional stock limits wait when their target is reached and resume after withdrawals. Crafted results return directly to MV.

Connect the Crafter directly through its network to MV Storage, or use an enabled bound MV Link through Transfer Pipes. Supply the Crafter with Watt power; remote MV input and output also require link transfer power. Missing materials or power cause a recoverable wait.

MV Storage keeps ordinary stackable items as counts. Use Find to filter stored names, including custom items. Shift-click deposits go into storage; deliberate grid placement sets the crafting recipe. MV Link has the same name-matching Find logic. Show all items resets its search; clicking an item selects its routing filter.

***JUMP TO:*** *Machine directory* | *Block recipes* | *Drop tables* | *Essence equipment recipes* | *Tree to Supremium costs*

# 11. Upgrade slots: speed, range and efficiency

| | | | |
|---|---|---|---|
|  *Speed Upgrade* |  *Range Upgrade* |  *Efficiency Upgrade* | |

Put real boards into compatible HUD slots. Silhouettes show supported kinds. Activators accept Speed only; Harvesters/Grinders accept Speed, Range and Efficiency; processors accept Speed and Efficiency.

The active cap is THREE boards TOTAL across the machine, not three per type. Stack duplicates in one slot or spread them across slots. Unsupported boards are rejected and excess boards are returned/ejected.

Speed produces x1 base, then x2/x3/x4 with one/two/three boards. Faster processor work increases its working power use. Each Range board adds two blocks to a supported working radius.

Each Efficiency board multiplies applicable working power cost by 0.8: 20%, 36% or about 49% saved before rounding. It does not reduce ingredients. Removing boards restores base behavior. Read live HUD stats and remove boards before breaking a protected machine.

***JUMP TO:*** *Machine directory* | *Block recipes* | *Drop tables* | *Essence equipment recipes* | *Tree to Supremium costs*

# 12. Planters, harvesters, sprinklers and Activators

| | | | |
|---|---|---|---|
|  |  |  |  |
| *Planter* | *Harvester* | *Automated Sprinkler* | *Activator* |
| Crafting recipe | Crafting recipe | Crafting recipe | Crafting recipe |

Provide a Planter with seeds/saplings, suitable soil, free space and power. Place the powered Harvester so its area covers mature crops/trees. Export its internal output with a transfer node. Tree harvesting has built-in crook behavior.

Use the Automated Sprinkler and its supported water/operation arrangement for growth assistance. Watering cans and bone meal provide manual help. Growing faster and harvesting faster are different problems.

Configure an Activator deliberately: face, aim, use/break click, sneaking, redstone and order. Tool durability is spent on the first usable tool one at a time, not randomly spread across tools.

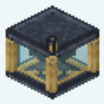
Ore crops have four harvest tiers. Inferium crops have five and always grow Inferium, with larger per-harvest yields. Check power and output capacity before adding speed boards.

***HUD EXAMPLES:*** *Activator: input tools, world action and returned output* | *Harvester: collected output and machine upgrades*

***BUILD EXAMPLES:*** *A small tree-harvesting platform* | *Crop growth with a water-fed sprinkler* | *An Activator aimed at a sapling*

***JUMP TO:*** *Machine directory* | *Block recipes* | *Drop tables* | *Essence equipment recipes* | *Tree to Supremium costs*

# 13. Processors and the Polymer factory

| | | | |
|---|---|---|---|
|  |  |  |  |
| *Pulverizer* | *Dehydrator* | *Glacial Chiller* | *Pressure Reactor* |
| Crafting recipe | Crafting recipe | Crafting recipe | Crafting recipe |

Put listed ingredients into INPUT, provide power/fluid as required, unpause and keep OUTPUT clear. Use the machine named on a card; a 3x3-looking inventory is not automatically a crafting bench.

Pulverizer makes its listed powders and crushed products. Dehydrator handles dry processing, Glacial Chiller handles cooling and water, and Magma Furnace provides its smelting route. Exact accepted inputs/outputs appear in the atlas.

Pressure Reactor batches use 16 accepted biomass items, Charcoal, water and power to make Polymer. Supported biomass includes its configured crops and saplings, not removed mulch. Mix accepted crop types when making a batch.

Polymer leads to Circuit Boards, upgrades and advanced factory components. Feed input and export finished output through nodes. Returned buckets need output space too. Verify a basic working line before adding speed.

***HUD EXAMPLES:*** *Pressure Reactor: inputs, water and Polymer output*

***BUILD EXAMPLES:*** *A first Polymer production cell*

***JUMP TO:*** *Machine directory* | *Block recipes* | *Drop tables* | *Essence equipment recipes* | *Tree to Supremium costs*

# 14. Advanced assembly, infusion and the Factory Core

| | | | |
|---|---|---|---|
|  |  |  | |
| *Precision Assembler* | *Material Infuser* | *Factory Core* | |
| Crafting recipe | Crafting recipe | Crafting recipe | |

Precision Assembler makes the listed Reinforced Frames, Advanced Circuits, Conductive Cores, Growth Essence and modifiers. Supply ingredients and Watts; the first complete matching recipe is processed.

Material Infuser needs solid ingredients plus water and power for Crystalline Alloy or Enriched Essence. Its fluid supply and returned containers are part of your production line.

Factory Core uses late components and substantial power for its listed high-throughput recipes. Build crop/Charcoal production, Polymer, circuits, assembly components and infusion materials in stages before scaling the core.

Keep buffers between stages. Use High-priority input destinations before bulk overflow. A blocked output or missing machine-made component can stop a large factory even when most materials are abundant.

***JUMP TO:*** *Machine directory* | *Block recipes* | *Drop tables* | *Essence equipment recipes* | *Tree to Supremium costs*

# 15. Build and run a seared smeltery

| | | | |
|---|---|---|---|
|  <br>*Seared Bricks*<br>Crafting recipe |  <br>*Smeltery Controller*<br>Crafting recipe |  <br>*Smeltery Drain*<br>Crafting recipe |  <br>*Seared Tank*<br>Crafting recipe |
|  <br>*Seared Faucet*<br>Crafting recipe | | | |

Build a 5 × 5 Seared Brick floor and a perimeter wall above it. The four outside corners of the floor and each wall layer are optional. Leave the center 3 × 3 open. Put the Controller, a Drain and a lava-filled Seared Tank in the wall. Place the Faucet outside the Drain and a Casting Table or Casting Basin immediately below the Faucet.

Put both metals in the Controller input. The smeltery melts one item, then advances to the next eligible slot. For example, copper in slot 1 and tin in slot 9 both get processed while the stacks stay where you placed them. The selected molten metal controls pouring. Choose the alloy recipe separately to mix it.

Supply lava through the connected Seared Tank arrangement. Insert supported solids in the controller, wait for melting, and select the metal you intend to inspect or pour. Distinct molten metals remain tracked separately.

Route a Drain/Faucet to the proper casting receiver below it. The Table handles reusable casts and parts; the Basin handles bulk block casting. Let a pour finish and cool before collecting.

Occupied/incompatible receivers stop a pour. Retrieve stored contents before rebuilding the structure. Powder, ingots and raw ore have different molten values; one powder is not one full ingot.

***JUMP TO:*** *Machine directory* | *Block recipes* | *Drop tables* | *Essence equipment recipes* | *Tree to Supremium costs*

# 16. Alloys, reusable casts and the Tool Forge



*Casting Table*

Crafting recipe



*Casting Basin*

Crafting recipe



*Tool Forge*

Crafting recipe



*Steel Ingot*

Make a reusable gold part cast by placing its stone pattern on the Table and pouring the listed Gold amount. Standard part-cast creation uses two Gold Ingots. Pour the desired metal into the cast, cool it and collect the part; the gold cast remains.

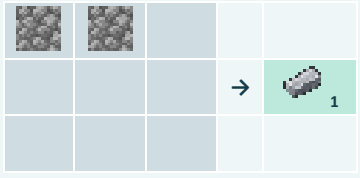
Alloy batches: 3 Copper + 1 Tin -> 4 Bronze; 1 Gold + 1 Silver -> 2 Electrum; 2 Iron + 1 Nickel -> 3 Invar. Steel uses 1 molten Iron plus 1 Coal and 1 Blaze Powder from solid input -> 1 Steel.

Select the desired alloy and enable Auto alloy. When its molten ingredients and any solid catalysts are ready, the controller makes the alloy. Select the finished metal to pour it into a suitable cast.

Cast a Head, Rod and Binding and assemble the supported modular pickaxe/hammer in the Tool Forge. Materials may differ between parts. The atlas includes the supported combinations.

Forge modifiers are not machine boards. Reinforcement Plate adds durability; Efficiency Module applies Efficiency II; Fortune Module applies Fortune I to a compatible pickaxe. Hold a modifier, sneak-use the Forge and select a tool. Restrictions on incompatible/equal-or-higher enchantments still apply.

**Stone Ingot Pattern**




Use the stone pattern on the Casting Table, then pour two ingots? worth of molten Gold to make a reusable Gold Ingot Cast.

***BUILD EXAMPLES:*** *Drain, faucet and casting receiver*

***JUMP TO:*** *Machine directory* | *Block recipes* | *Drop tables* | *Essence equipment recipes* | *Tree to Supremium costs*

# 17. Tool roles, metal armor and vein mining

| | | | |
|---|---|---|---|
|  |  |  |  |
| *Supremium Pickaxe* | *Supremium Crook* | *Supremium Hammer* | *Supremium Paxel* |

Pickaxes mine stone/ores; axes mine wood and strip logs; shovels dig soil and make paths; hoes till suitable soil. Paxels combine pickaxe/axe/shovel mining and applicable stripping/path actions.

Crooks use leaf bonus rolls. Hammers crush supported blocks. Sickles and scythes harvest nearby mature crops, with the scythe covering the larger area. Swords provide combat damage. Tool wear and enchantment compatibility depend on the tool.

Sneak-mine with a supported pickaxe, axe, shovel, paxel, hammer or crook to mine connected matching blocks. Cap: 64 total, or earlier when a finite tool breaks. Release Sneak or switch tools to stop queued work.

Crook leaf rolls and hammer conversions happen per block in a vein. Unbreakable Supremium tools still obey the 64-block cap. Fortune is not a universal multiplier for every scripted crushing or machine output.

Ordinary metal armor uses normal crafting-table armor shapes with its own ingots. Modular tools use the Forge. Essence gear uses a different tier-upgrade route in the next chapter.

***JUMP TO:*** *Machine directory* | *Block recipes* | *Drop tables* | *Essence equipment recipes* | *Tree to Supremium costs*

# 18. Essence armor: the complete five-tier progression

| | | | |
|---|---|---|---|
|  |  |  |  |
| *Inferium Essence* | *Infusion Crystal* | *Essence Tinkering Table* | *Supremium Chestplate* |
| | | Crafting recipe | |

Sieve Dirt for Inferium Essence and Sand for Prosperity Shards. Make an Essence Crafting Seed with one Wheat Seed and four Prosperity Shards in a cross. Surround that seed with four Inferium Essence in the same cross. Plant the resulting Inferium Essence Seeds and interact with mature plants to harvest without removing the seed.

Craft the Infusion Crystal at a Crafting Table using the pictured recipe in the Essence appendix. Then put the finished crystal into the Essence Tinkering Table.

Craft the Essence Tinkering Table at a Crafting Table. Follow its pictured recipe. Place and open it. It runs without Watt power.

Insert four lower essence plus the already-crafted crystal into its input slots. Unpause and wait four seconds. It batches up to 64 output and retains the crystal, spending one use per infused output. Chain: Inferium -> Prudentium -> Intermedium -> Superium -> Supremium, always 4:1.

The crystal lasts 1,000 infused outputs. A Master Crystal has unlimited uses: normal crystal in the center, four Prosperity Ingots on the edges and four Supremium Essence in the corners.

Craft ingots at a normal table: Iron Ingot + four Prosperity Shards -> Prosperity Ingot. Prosperity Ingot + four Inferium Essence -> Inferium Ingot. Each later ingot uses one previous-tier ingot + four matching next-tier essence. All are shapeless.

Craft initial Inferium armor in normal armor shapes: helmet 5, chestplate 8, leggings 7, boots 4; 24 Inferium Ingots total. Upgrade one previous armor piece + four NEXT-tier ingots in the Essence Tinkering Table. Every complete set upgrade uses 16 next-tier ingots.

Full-set bonuses: Inferium is base armor; Prudentium gives water breathing; Intermedium gives step assist; Superium prevents fall damage; Supremium prevents fall damage and grants native flight. Wear all four matching pieces. Higher sets do not automatically inherit all earlier bonuses. Supremium gear is unbreakable.

For all armor through Supremium, plan for 24 Inferium Ingots, then 16 each Prudentium/Intermedium/Superium/Supremium Ingots. Make EXTRA lower-tier ingots for those higher ingots: ingots already consumed by armor cannot be reused. Gear upgrades preserve names, enchantments and proportion of durability used.

Upgrade seed tiers at a crafting table with the previous seed plus four of the next essence in a cross. They yield 1/2/3/4/5 INFERIUM per harvest, never higher essence directly. Build the farm first, then infuse its output for gear.

Native flight requires Education features and Cheats. Equip the full Supremium set in Survival/Adventure, wait for ready, then double-tap Jump. Jump rises, Sneak descends, sprint boosts speed; double-tap stops flight. No flight charm is needed. Removing a piece revokes and resets active flight, restoring the original Survival/Adventure mode. Creative and Spectator flight are left alone.

***PICTURED RECIPES*** *Infusion Crystal* | *Essence Tinkering Table*

***HUD EXAMPLES:*** *Essence Table: crystal and essence input*

***BUILD EXAMPLES:*** *Essence infusion without power*

***JUMP TO:*** *Machine directory* | *Block recipes* | *Drop tables* | *Essence equipment recipes* | *Tree to Supremium costs*

# 19. Essence charms and watering cans

| | | | |
|---|---|---|---|
|  |  |  |  |
| *Blank Essence Charm* | *Speed Essence Charm* | *Inferium Watering Can* | *Supremium Watering Can* |

Blank Essence Charm: one Prosperity Ingot + four Supremium Essence. A finished charm uses the blank, two Supremium Essence and its specialty ingredient. Put it with a compatible Supremium piece in the Essence Table to install one charm. Charmed gear plus Flint removes and returns the charm.

Armor charms include Speed, Night Vision, Water Breathing, Fire Resistance, Resistance and No Fall. Tool charms include compatible Strength, Fortune, Silk Touch, Efficiency and Mining Area effects. One charm per item. See the atlas for every compatible target and ingredient.

Craft an Inferium Watering Can from its ingots and a Bucket. Later can recipes use the previous can. Use on water to fill, then on a crop/soil to water the nearby area. Higher tiers have larger areas.

Remaining water percentage appears while held, with exact water/capacity in the tooltip. An accepted watering action spends one unit immediately, even on mature crops. The first successful plant growth is included; additional plants spend another unit each. Refill when empty.

A short hand stream and target splashes animate watering. Effects are throttled and spend no extra water. Switching the selected slot cancels remaining queued work. Watering a mature plant does not create an additional harvest.

***JUMP TO:*** *Machine directory* | *Block recipes* | *Drop tables* | *Essence equipment recipes* | *Tree to Supremium costs*

# 20. Mobs, Safari Nets, meat, slime and Bacon

| | | | |
|---|---|---|---|
|  <br>*Safari Net* |  <br>*Mob Grinder*<br>Crafting recipe |  <br>*Slaughter Factory*<br>Crafting recipe |  <br>*Mob Duplicator*<br>Crafting recipe |
|  <br>*Bacon Dispenser* | | | |

Hostile mobs need suitable darkness, full-block spawning space, headroom and distance from the player. Passive animals need suitable grass, light, space and their spawning conditions. A platform directly under where you stand is not an adequate spawning test. For a practical test, stand about 30 blocks from the platform while keeping it inside the loaded simulation area. Passive spawning can take several minutes and depends on biome and existing populations.

A Safari Net captures a supported mob for safe release or the applicable machine route. It is obtainable in Survival. Keep valuable named mobs out of initial machine tests and provide safe release space.

Face the Mob Grinder's blades toward the mobs. The HUD names the direction and range it checks. Base cost is 80 W per mob. Transfer Nodes can extract item drops, and separate tank groups can collect Liquid XP and Pink Essence. Full fluid buffers preserve their stored contents and skip new overflow. Full item output still stops processing.

The Pink Slime Generator turns late mob production into power. Raw/cooked meat block packing and unpacking routes appear in the atlas.

The Bacon Dispenser stores more than one visual stack. Load while carrying Bacon using the supported use action and read its stored status. Its automatic feeding works in supported non-Creative modes when hungry. A native single-stack preview is not the authoritative stored total.

***JUMP TO:*** *Machine directory* | *Block recipes* | *Drop tables* | *Essence equipment recipes* | *Tree to Supremium costs*

# 21. Builder's Wand and early flight equipment

| | | | |
|---|---|---|---|
|  *Builder's Wand* |  *Jetpack* | | |

Hold the Builder’s Wand and Sneak + use to open its block selector. Choose a building block available in your inventory or MV stock, and choose Fill or Replace and Box or Dome. Close the HUD, aim at a block within 32 blocks, and left-click to select the start. Aim to resize the preview and left-click again to begin building. Sneak + left-click cancels the selection or active build. Supply the selected blocks; the wand is not a free-block generator.

Jetpacks are chest equipment for earlier held-Jump lift. Equip one and hold Jump to rise; release Jump or Sneak to stop its lift. Flight consumes durability over time, and the pack stops when it wears out. This is a different control scheme from the full Supremium set’s native flight. Test above a safe platform and do not assume a jetpack provides the full armor set’s fall protection.

For Void platform mode, select the first corner under your feet, move or fly to the second corner, then left-click to build a flat platform between them. In other modes, Sneak + left-click cancels the selection or active build. Both use the selected blocks from your inventory or MV stock.

***JUMP TO:*** *Machine directory* | *Block recipes* | *Drop tables* | *Essence equipment recipes* | *Tree to Supremium costs*

# 22. Scale the factory and finish progression

| | | | |
|---|---|---|---|
|  |  |  | |
| *Factory Core* | *Black Hole Unit* | *Battery Bank* | |
| Crafting recipe | Crafting recipe | Crafting recipe | |

Build stable loops in stages. Start with trees, Cobblestone and sieve outputs. Make early metals and power, then build the smeltery. Make Steel and Netherite before attempting MV Storage. Expand with Polymer, automated crops and advanced processing.

Use the optional achievements checklist to track milestones. Keep a working supply of intermediate ingredients for each new production tier.

Buffer intermediate products and prioritize active production before bulk overflow. Monitor whether a bottleneck is farm growth, power, a wrong filter, loaded chunks or full output.

Batteries protect outages but do not replace sufficient primary generation. Increase sources, input and extraction together. Fewer loose item entities and purposeful layouts can perform better than many redundant networks.

***JUMP TO:*** *Machine directory* | *Block recipes* | *Drop tables* | *Essence equipment recipes* | *Tree to Supremium costs*

# 23. Troubleshooting and safe pack updates

| | | | |
|---|---|---|---|
|  |  |  | |
| *Essence Tinkering Table* | *Infusion Crystal* | *MV System* | |
| Crafting recipe | | Crafting recipe | |

Essence Table idle: craft the crystal at a normal Crafting Table first. Then insert essence + crystal, previous gear + next-tier ingots, or Supremium gear + charm. Its input grid is not a general crafting bench. It needs no Watt Pipe.

If native flight persists after removing Supremium armor, verify that the world is using matching current Behavior and Resource Packs and reload. Record the Bedrock version when reporting a persistent issue.

No essence from a few dirt blocks: it is a chance drop, not guaranteed. The six-click flow is tested at all mesh tiers. Process more dirt and collect physical drops safely.

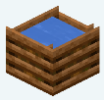
When mined, supported machines with stored items can drop a Packed Machine. Place it to restore its contents. Empty machines drop normally. MV Links, Composters and barrels are excluded from packing. Leave hidden storage hosts intact.

Wrong transfers: check intake direction, active ALLOW/BLOCK mode, saved filter items, priority, MV Link direction, power, loaded storage and free destination space.

Two pack versions: an older embedded world pack can coexist with a new imported library version. Check BP and RP descriptions, then reload Minecraft. You do not have to delete the world.

***JUMP TO:*** *Machine directory* | *Block recipes* | *Drop tables* | *Essence equipment recipes* | *Tree to Supremium costs*

# 24. Material recipes: End Stone and other progression blocks

| | | | |
|---|---|---|---|
|  <br>*Stone Barrel*<br>Crafting recipe |  <br>*Fluid Pump*<br>Crafting recipe |  <br>*Moisture Collector*<br>Crafting recipe |  <br>*Rainwater Composter* |
|  <br>*Ancient Spores* | | | |

END STONE: place a Stone Barrel. Supply at least 100 mB of lava, then insert 1 Glowstone Dust to produce 1 End Stone. A lava bucket holds 1,000 mB and supports ten conversions. For automation, connect a lava tank through the fluid network and insert Glowstone Dust through a Transfer Node or MV Link. Extract End Stone from the barrel output. No Watt power is needed by the barrel itself.

NETHERRACK: use the same Stone Barrel with 100 mB lava plus 1 Redstone for 1 Netherrack. A hammer breaks Netherrack into Crushed Netherrack; sieve the crushed material for Nether resources. Consult the linked sieve tables for exact rolls and mesh multipliers.

OBSIDIAN: insert 1 Water Bucket into a Stone Barrel containing at least 100 mB lava. This pack produces 1 Obsidian and returns the empty bucket. A Clay Water Bucket also works and returns a Clay Bucket. These are pack-specific conversion costs.

MYCELIUM: obtain Ancient Spores by sieving the supported sand inputs, then use Ancient Spores on Dirt. Mycelium is required below the water-filled composter for Witch Water. See the sieve tables for the current chance for your input and mesh.

SOUL SAND. Put a full ordinary-water Composter above Mycelium and wait for Witch Water. Use one Sand to make one Soul Sand, then refill and wait for the next batch. Bring water by bucket or use four ordinary leaf blocks in the empty Composter.

CLAY AND GROUT. Hammer Sand into Dust, then use one Dust in a full ordinary-water Composter for one Clay Block. Break it into Clay Balls. Craft one Clay Ball, one Sand and one Gravel into two Grout. Smelt Grout into Seared Brick in a regular Furnace or Magma Furnace.

STEEL: build a valid smeltery and provide lava fuel. The Steel alloy uses 1 Iron + 1 Coal + 1 Blaze Powder to produce 1 Steel. Select the stored metal to cast it through a Drain and Faucet into a suitable cast. Selecting Steel does not itself supply missing alloy ingredients.

NETHERITE. Make Netherrack from lava and Redstone Dust. Hammer it into Crushed Netherrack and sieve it for Netherite Scrap. The Basic mesh gives 1%, Iron 1.25%, Diamond 1.6% and Netherite 2% per input. Combine four Netherite Scrap and four Gold Ingots into one Netherite Ingot.

ESSENCE EQUIPMENT: start with Prosperity Shards and Inferium from the documented sieve/crop routes. Grow essence crops to maturity before harvesting. Use the infusion progression and tiered ingot recipes in the complete essence recipe section, then follow each armor or tool recipe. The seed is not itself an ingot; repeated mature harvests supply its ingredients.

***BUILD EXAMPLES:*** *End Stone: Glowstone Dust and lava*

***JUMP TO:*** *Machine directory* | *Block recipes* | *Drop tables* | *Essence equipment recipes* | *Tree to Supremium costs*

# Custom HUDs: inputs, outputs and controls

These are annotated reference schematics based on the current HUD controls and slot roles, not live screenshots. Representative input and output stages may be shown together to explain a recipe. Block placement is covered in the linked build examples.

***Pressure Reactor: inputs, water and Polymer output*** *Open annotated HUD*

***Essence Table: crystal and essence input*** *Open annotated HUD*

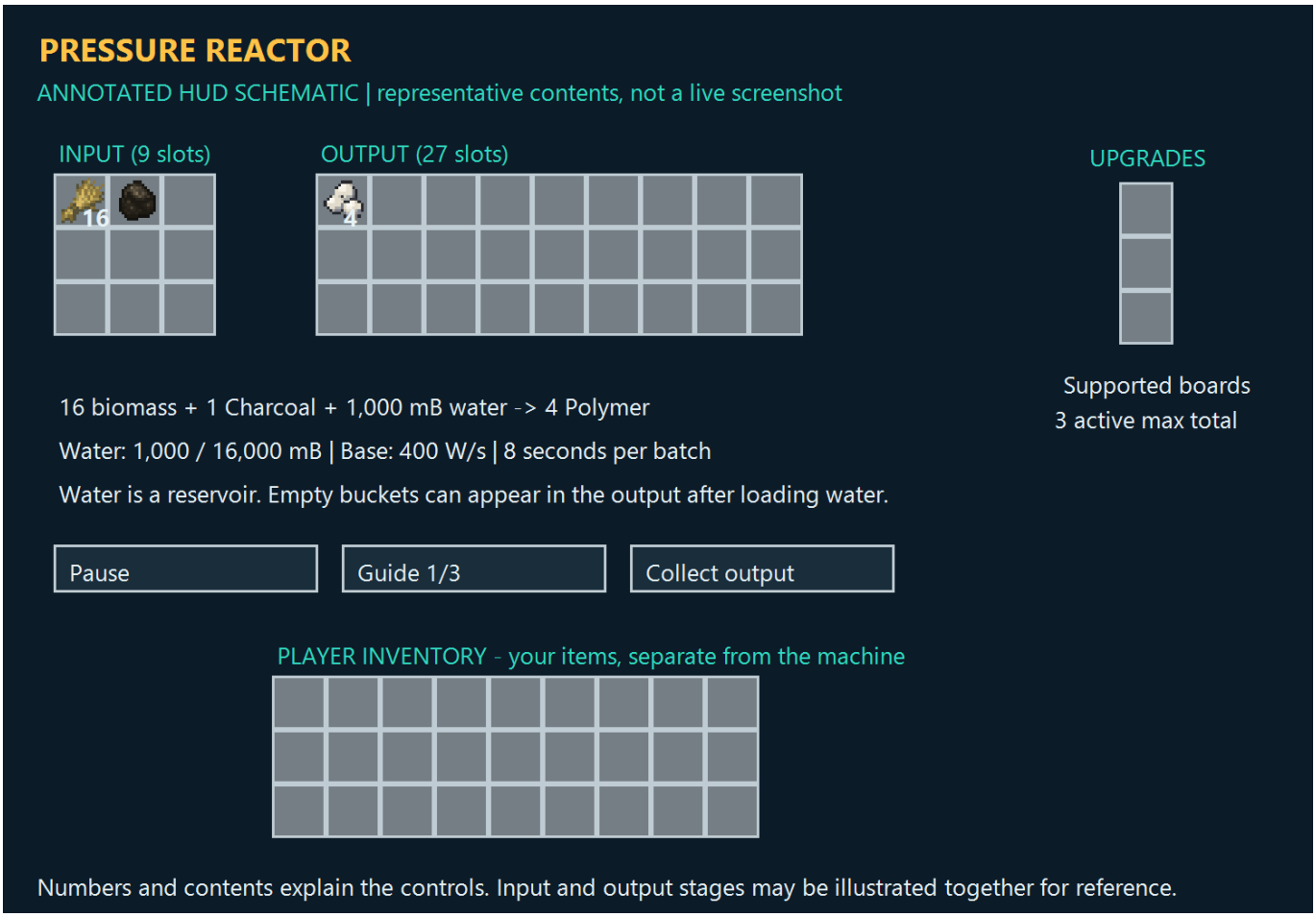
***Smart Transfer Node: filters are not output slots*** *Open annotated HUD*

***Activator: input tools, world action and returned output*** *Open annotated HUD*

***MV Storage: storage totals, recipe pattern and player inventory*** *Open annotated HUD*

***Harvester: collected output and machine upgrades*** *Open annotated HUD*

## Pressure Reactor: inputs, water and Polymer output



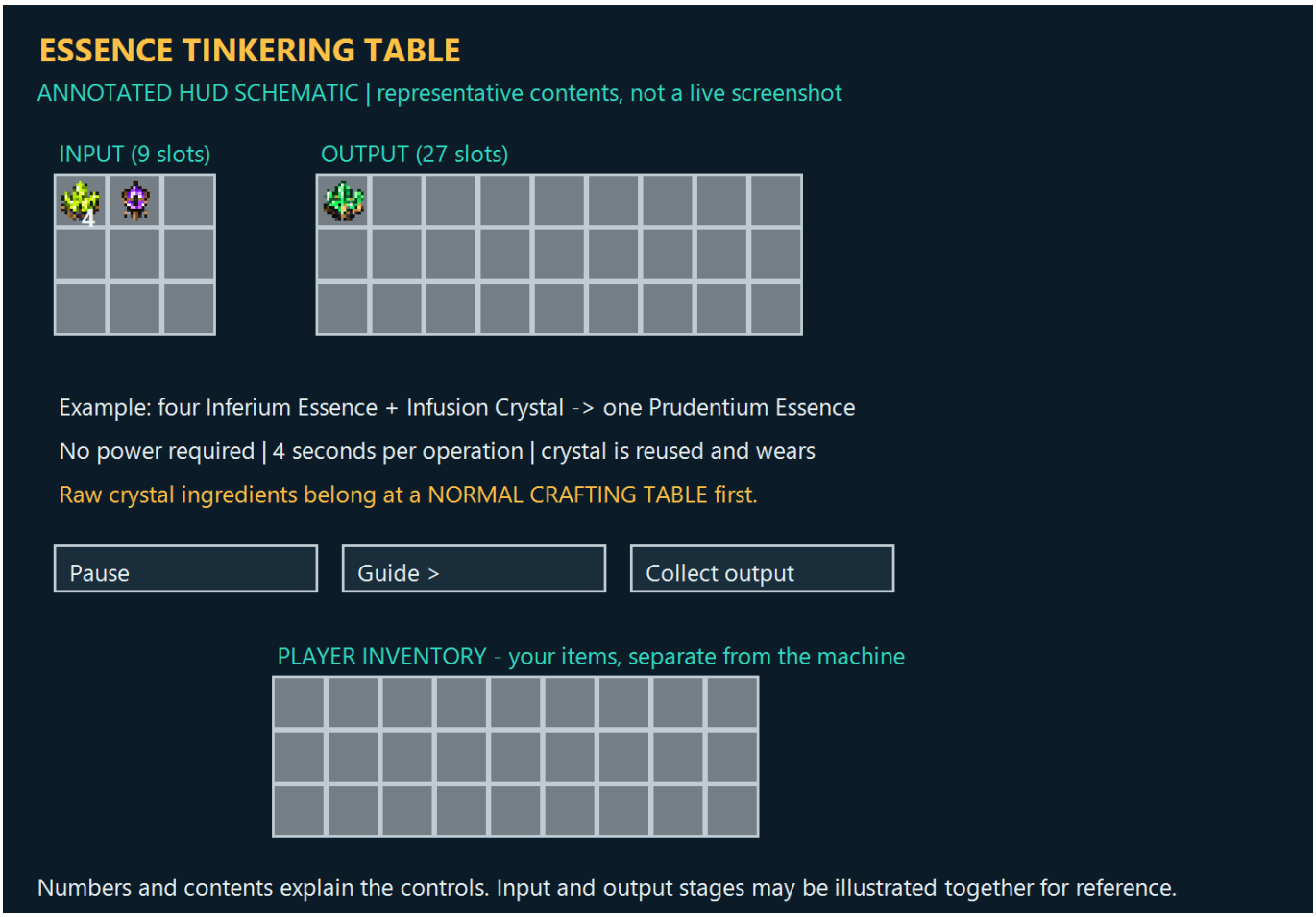

**1.** Put biomass and Charcoal in the nine INPUT slots. The example uses 16 Wheat and 1 Charcoal. Supply 1,000 mB water in the machine reservoir.

**2.** The 27 OUTPUT slots hold finished Polymer and returned empty buckets. The player inventory below is separate storage; items left there are not automatically processor inputs.

**3.** Pause stops the job. Guide cycles recipe help. Collect output moves finished items to your inventory. Use only supported boards in the upgrade slots, with three active upgrades total.

***BUILD THIS:*** *Placement example* | *Processors and the Polymer factory* | *All HUD examples*

## Essence Table: crystal and essence input



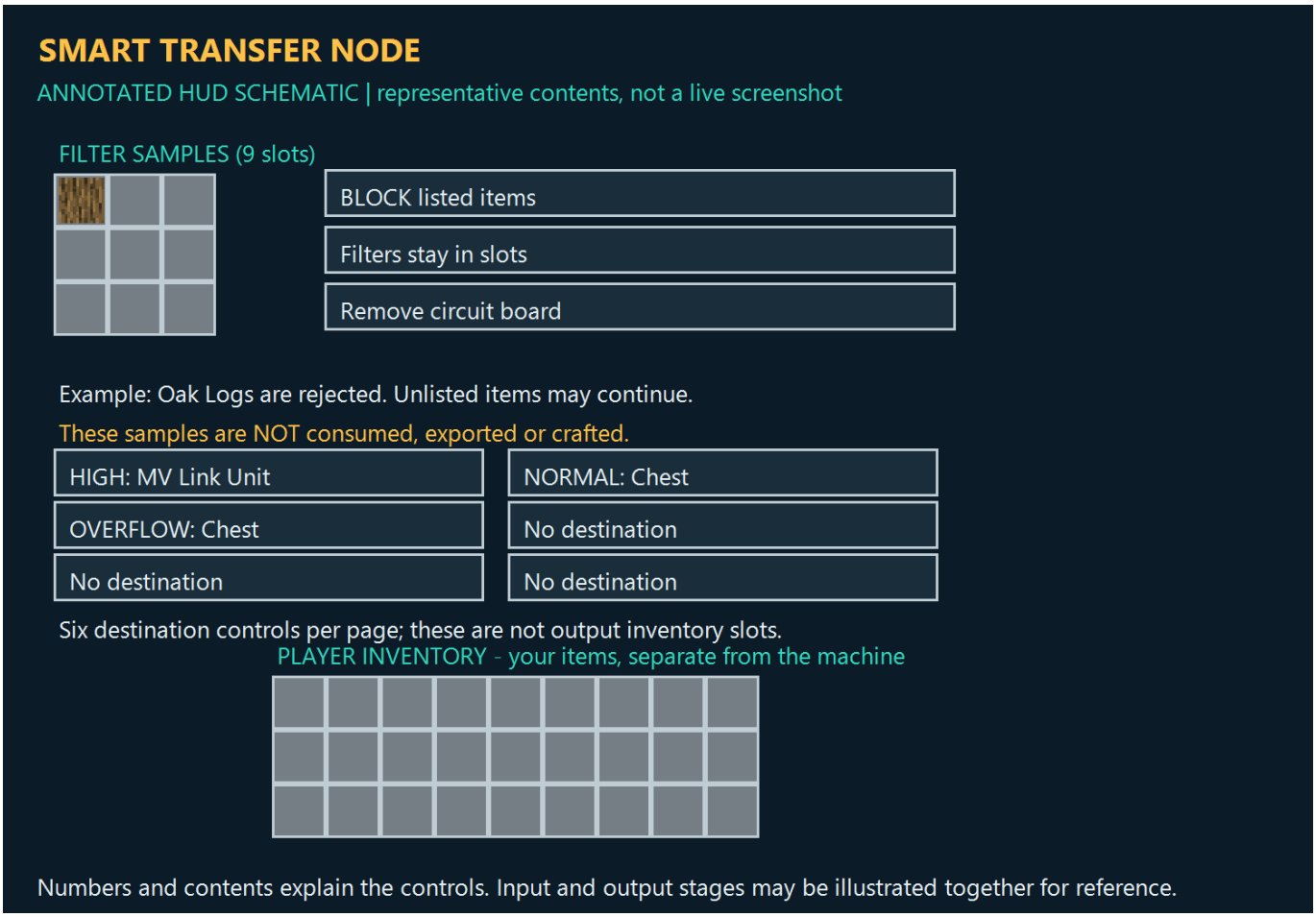

**1.** Put four lower-tier essence and the real Infusion Crystal in INPUT. One operation produces one next-tier essence in OUTPUT. The crystal remains available and spends durability.

**2.** This machine does not need power. Make the raw Infusion Crystal recipe at a normal Crafting Table first. Raw Gold, Diamond and Prosperity Shards do not perform infusion.

**3.** The same table also supports eligible equipment upgrades and charms. Use the named recipe requirements; an item picture alone does not make it a valid ingredient.

***BUILD THIS:*** *Placement example* | *Essence armor: the complete five-tier progression* | *All HUD examples*

## Smart Transfer Node: filters are not output slots



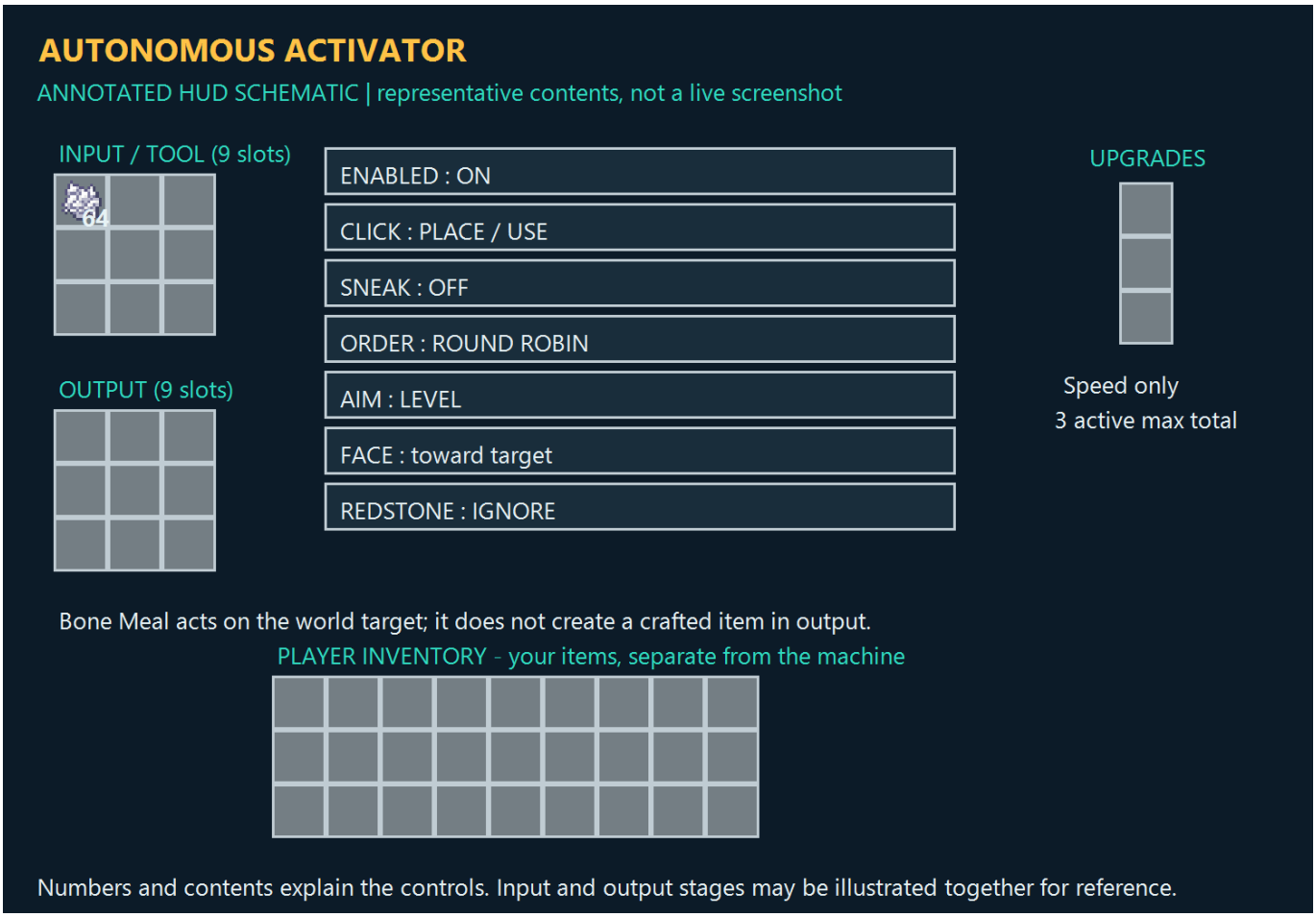

**1.** The 3 x 3 filter area holds real sample items and keeps them until you remove them. They are not crafting ingredients or exported products.

**2.** Click the mode control to choose ALLOW or BLOCK. In this example, BLOCK plus an Oak Log sample rejects Oak Logs while unlisted items remain eligible.

**3.** The six destination controls set routing priorities. HIGH is tried before NORMAL, then OVERFLOW. They are buttons, not inventories. The node still needs a correctly aimed source intake and a connected item route.

***BUILD THIS:*** *Placement example* | *Transfer Nodes, filters, priorities and bulk storage* | *All HUD examples*

## Activator: input tools, world action and returned output



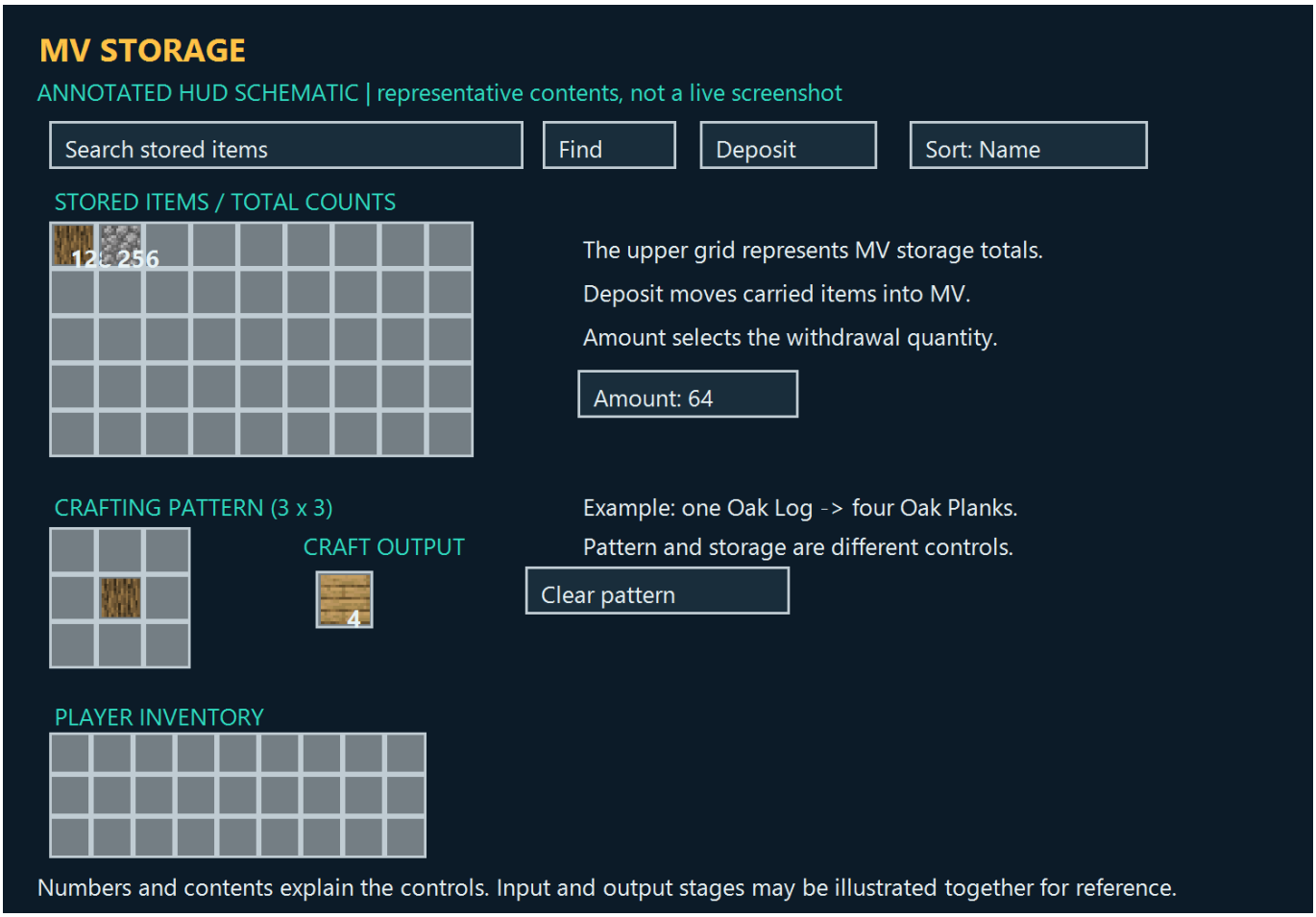

**1.** Put Bone Meal, tools or other supported usable items in INPUT / TOOL. Enable the machine, choose the click mode, and set its face and aim toward the target.

**2.** The OUTPUT slots hold any returned or produced items from supported actions. Bone Meal growth changes the target in the world; a crafted item is not expected in OUTPUT.

**3.** Tool durability is spent one tool at a time in slot order. This machine supports Speed upgrades only. Removing boards restores the normal action interval.

***BUILD THIS:*** *Placement example* | *Planters, harvesters, sprinklers and Activators* | *All HUD examples*

## MV Storage: storage totals, recipe pattern and player inventory



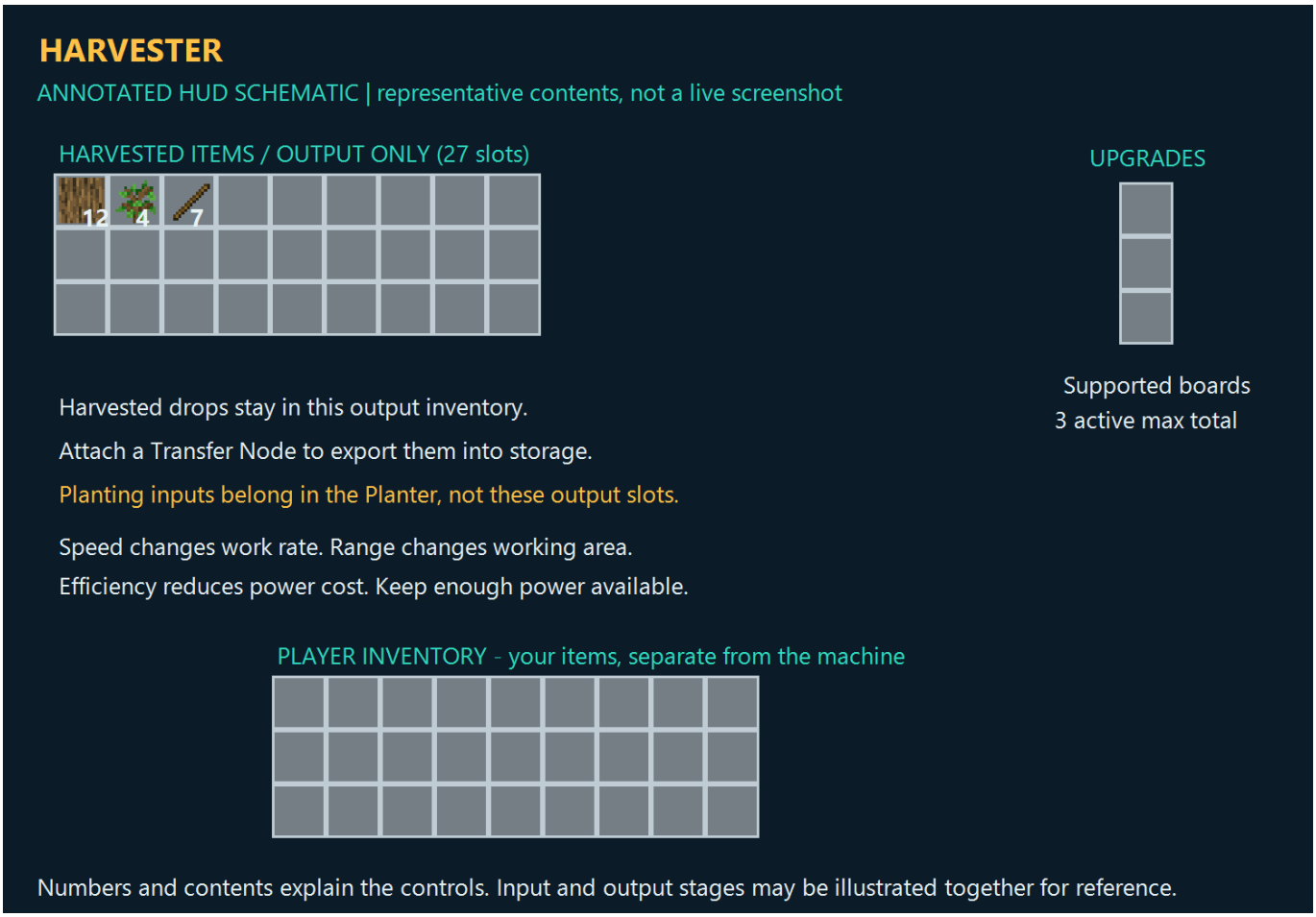

**1.** Search and Find act on stored items. Deposit sends carried items into MV. The upper grid shows storage contents and totals; the player inventory at the bottom is separate.

**2.** Amount selects the withdrawal quantity. Use the integrated 3 x 3 crafting pattern to choose a recipe and the output control to craft when the required ingredients are available.

**3.** The example illustrates Oak Log to Oak Planks. Clear empties the pattern selection. The handheld remote opens this same storage interface after binding to the powered MV system.

***BUILD THIS:*** *Placement example* | *MV Storage, in-screen crafting and wireless access* | *All HUD examples*

## Harvester: collected output and machine upgrades




**1.** The Harvester grid is OUTPUT storage for collected drops. Do not treat it as a planting-input inventory. Seeds and saplings for planting belong in the Planter.

**2.** Attach a Transfer Node aimed at the Harvester to export stored drops. A full output can stop useful work; check extraction before adding more Speed upgrades.

**3.** Speed changes work rate, Range changes the working area, and Efficiency reduces the power cost. The supported upgrade slots share a maximum of three active upgrades total.

***BUILD THIS:*** *Placement example* | *Planters, harvesters, sprinklers and Activators* | *All HUD examples*

# Illustrated build examples

Choose a build. These offline 3D renders place the pack models on one shared block grid, with a common camera and depth buffer. Numbered markers identify parts; the steps below each picture explain placement, supplies, settings and output. They are reference renders, not in-game screenshots. Every example links back to its system chapter and relevant crafting recipes.

***Eight saplings into dirt*** *Open illustrated steps*

***Water collector, pump and tank*** *Open illustrated steps*

***Witch Water into Soul Sand*** *Open illustrated steps*

***Water conversions: clay and mud*** *Open illustrated steps*

***A compact sieve workshop*** *Open illustrated steps*

***A small tree-harvesting platform*** *Open illustrated steps*

***Crop growth with a water-fed sprinkler*** *Open illustrated steps*

***A battery bank on a shared power line*** *Open illustrated steps*

***Drain, faucet and casting receiver*** *Open illustrated steps*

***Filtered chest-to-chest transport*** *Open illustrated steps*

***MV Storage with a handheld remote*** *Open illustrated steps*

***A first Polymer production cell*** *Open illustrated steps*

***Essence infusion without power*** *Open illustrated steps*

***An Activator aimed at a sapling*** *Open illustrated steps*

***Bulk storage off a farm line*** *Open illustrated steps*

***End Stone: Glowstone Dust and lava*** *Open illustrated steps*

## Eight saplings into dirt




***INPUT:*** *8 saplings*

***OUTPUT:*** *1 Dirt*

**1.** Place an empty composter. Keep water out of this batch.

**2.** Use eight saplings on the empty Composter. All eight fill steps are green.

**3.** Wait for the full batch to compost into dirt, then collect it before refilling.

***READ MORE:*** *Composting, water, clay and mycelium* | *All build examples*

***CRAFT THE BLOCKS:*** *Moisture Collector* | *Sieve*

## Water collector, pump and tank




***INPUT:*** *Lava for the Moisture Collector*

***OUTPUT:*** *Water stored in connected tanks*

**1.** Place a Moisture Collector on solid ground, with an empty Composter directly above it.

**2.** Use a Lava Bucket on the collector or connect a lava Tank. The collector adds water to the Composter as long as it has lava and there is space.

**3.** Place the Fluid Pump against the water source and run its purple Transfer Pipe route into matching tanks.

**4.** Watch the source and tank totals. A full or incompatible destination must wait without deleting source water.

***READ MORE:*** *Fluid Pumps, tanks and lava handling* | *All build examples*

***CRAFT THE BLOCKS:*** *Fluid Pump* | *Tank* | *Transfer Pipe*

## Witch Water into Soul Sand




***INPUT:*** *Mycelium + composter + 1,000 mB water + 1 Sand*

***OUTPUT:*** *1 Soul Sand per conversion*

**1.** Sieve Sand or Red Sand for Ancient Spores. Use the spores on Dirt to make Mycelium.

**2.** Place the composter directly on Mycelium. Fill it with one bucket of ordinary water (1,000 mB).

**3.** Keep the area loaded for 30 one-second processing ticks. Its status changes to Witch Water. Do not connect an ordinary-water extraction route.

**4.** Use one Sand on the ready Witch Water. Collect the Soul Sand, refill the water, and repeat.

***READ MORE:*** *Composting, water, clay and mycelium* | *All build examples*

***CRAFT THE BLOCKS:*** *Sieve*

***DROP CHANCES:*** *All inputs and meshes*

## Water conversions: clay and mud




***INPUT:*** *1,000 mB ordinary water + Dust OR Dirt*

***OUTPUT:*** *Clay OR Mud*

**1.** Fill an empty Composter with four ordinary leaf blocks or one Water Bucket. Set it on ordinary ground for these recipes.

**2.** For clay, use one Dust on the water. For mud, use one Dirt.

**3.** Collect the finished result before changing recipes. Refill one bucket of water for each new conversion.

***READ MORE:*** *Composting, water, clay and mycelium* | *All build examples*

***CRAFT THE BLOCKS:*** *Stone Barrel*

## A compact sieve workshop




***INPUT:*** *One sieve input + installed mesh*

***OUTPUT:*** *Resource drops listed by input and mesh*

**1.** Use the supported hammer chain: Cobblestone to Gravel to Sand to Dust. Choose the stage you need from the drop tables.

**2.** Install a mesh in the Sieve. Load one supported input and work it through six clicks.

**3.** Keep room for ejected drops. A nearby Vacuum Hopper is a collection example; add a Transfer Node if you want automatic export to the chest.

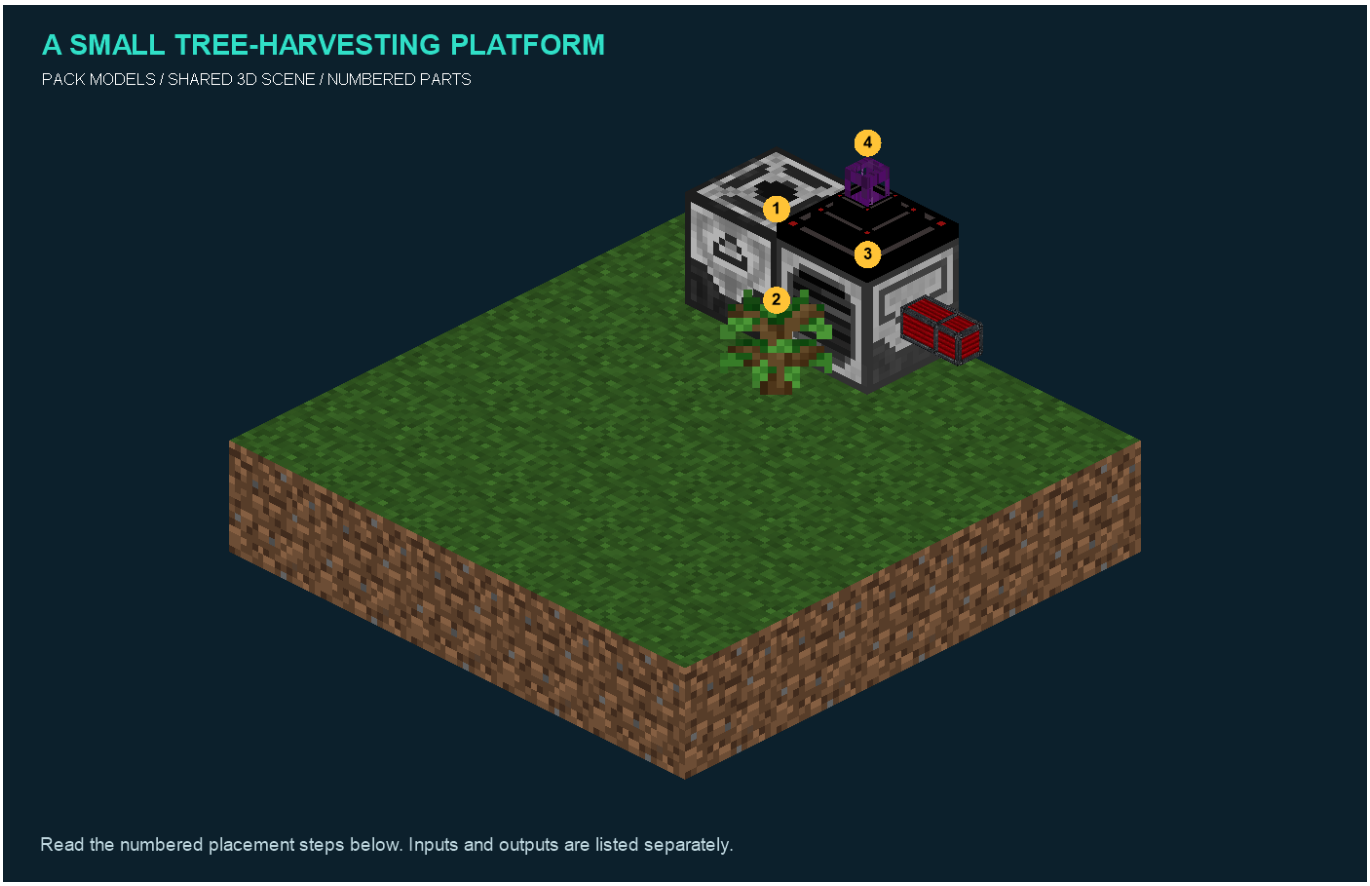
**4.** Upgraded meshes change the independent drop rolls. Follow the linked drop tables for exact chances and quantities.

***READ MORE:*** *Hammers, sieving and mesh progression* | *All build examples*

***CRAFT THE BLOCKS:*** *Sieve* | *Vacuum Hopper*

***DROP CHANCES:*** *All inputs and meshes*

## A small tree-harvesting platform




***INPUT:*** *Valid soil, supported planting inputs and power*

***OUTPUT:*** *Harvested wood, leaves and tree drops*

**1.** Build a soil platform with open space above and around the tree. The drawing is a small starter example; leave clearance for the species you grow.

**2.** Place the Planter next to the Harvester and supply supported seeds or saplings. This example keeps the tree one block in front of the Harvester, inside its unmodified range.

**3.** Place the powered Harvester within working range. Its default working area and upgrade range must cover the tree.

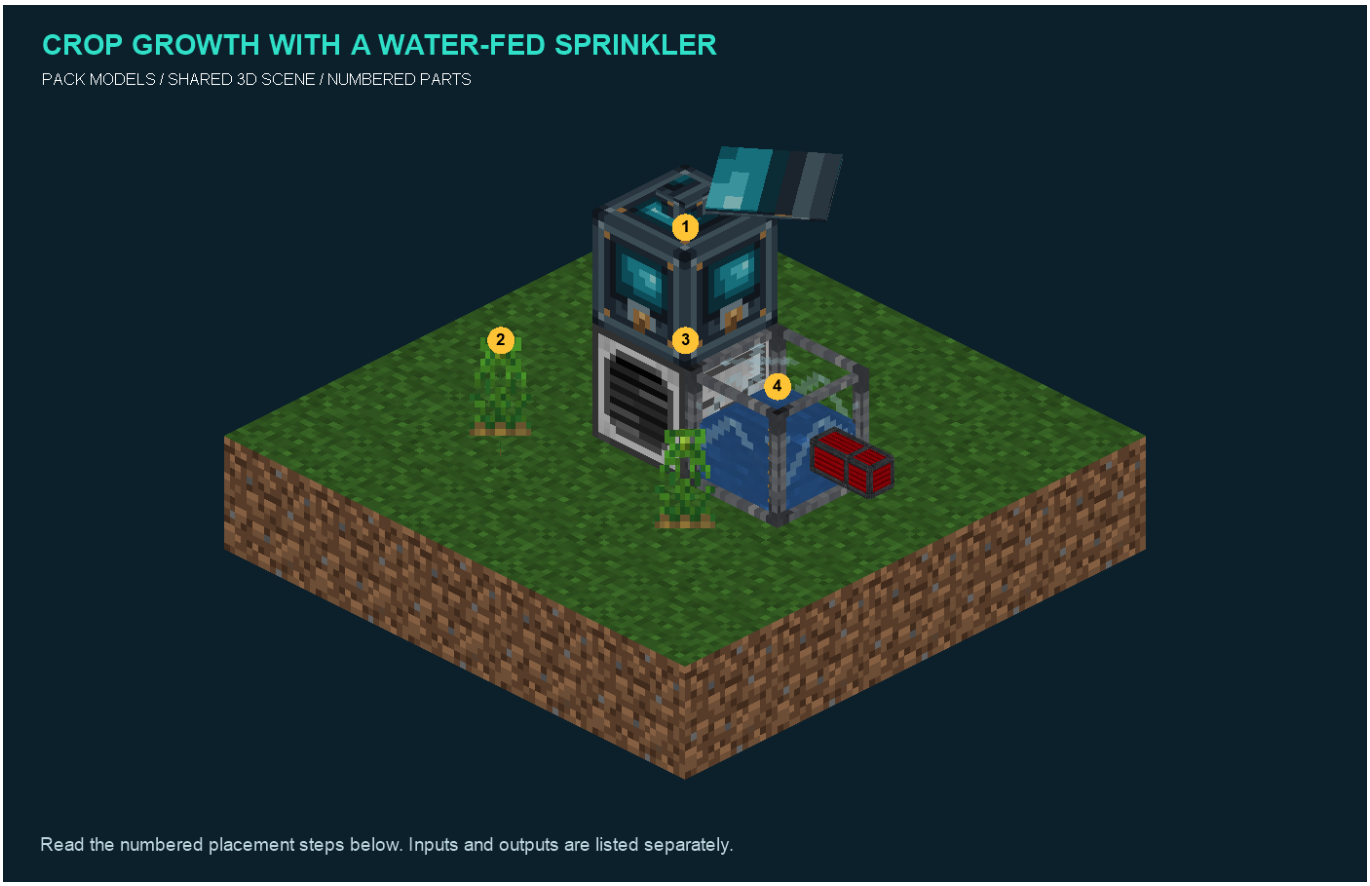
**4.** Attach an extraction node to the Harvester output and export drops. Keep a separate seed return supply for the Planter.

***READ MORE:*** *Planters, harvesters, sprinklers and Activators* | *All build examples*

***HUD:*** *Inputs, outputs and controls*

***CRAFT THE BLOCKS:*** *Planter* | *Harvester* | *Transfer Node*

## Crop growth with a water-fed sprinkler




***INPUT:*** *Growing crops + water + power*

***OUTPUT:*** *Faster growth; separate machine harvest*

**1.** Build the soil platform and plant supported crops. Center the Harvester and place the sprinkler one block above it, so the surrounding plants are within both machines' unmodified horizontal range.

**2.** Keep growing plants within the sprinkler's 5 x 5 candidate area, including its supported height band.

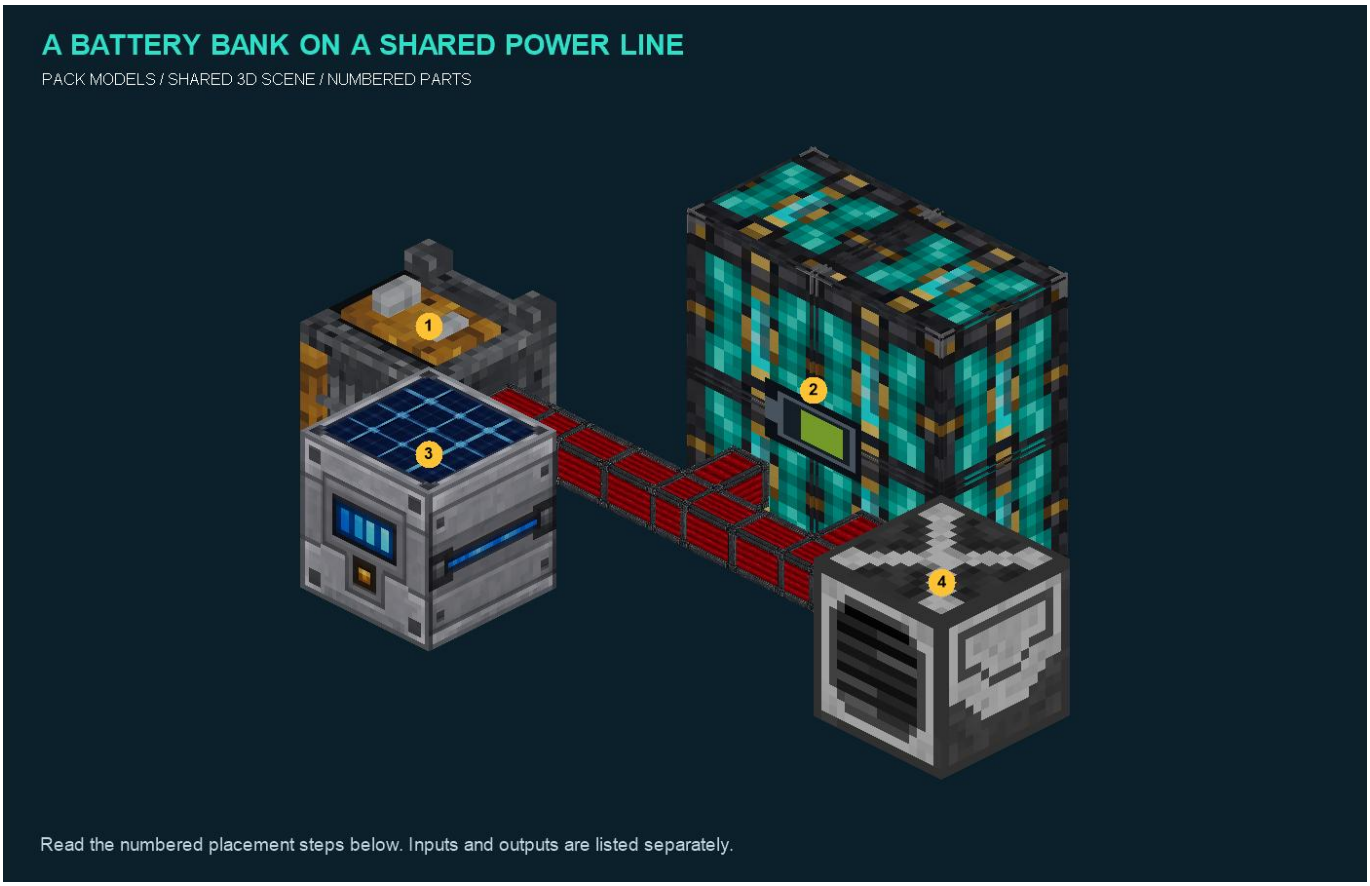
**3.** Supply water and power. A working sprinkler pulse needs 25 mB water and 10 W before efficiency changes; empty supplies pause it.

**4.** The sprinkler grows crops; a separate Harvester collects mature crops and a Planter replenishes planting inputs.

***READ MORE:*** *Planters, harvesters, sprinklers and Activators* | *All build examples*

***CRAFT THE BLOCKS:*** *Automated Sprinkler* | *Harvester* | *Planter*

## A battery bank on a shared power line




***INPUT:*** *Active primary power sources*

***OUTPUT:*** *Stored backup energy when primary generation stops*

**1.** Connect the generator and machine to the same Watt Pipe network. Batteries are a connected branch, not a mandatory pass-through.

**2.** Place batteries adjacent to combine storage capacity. Add more active sources, such as solar, to the same network to increase available charging power.

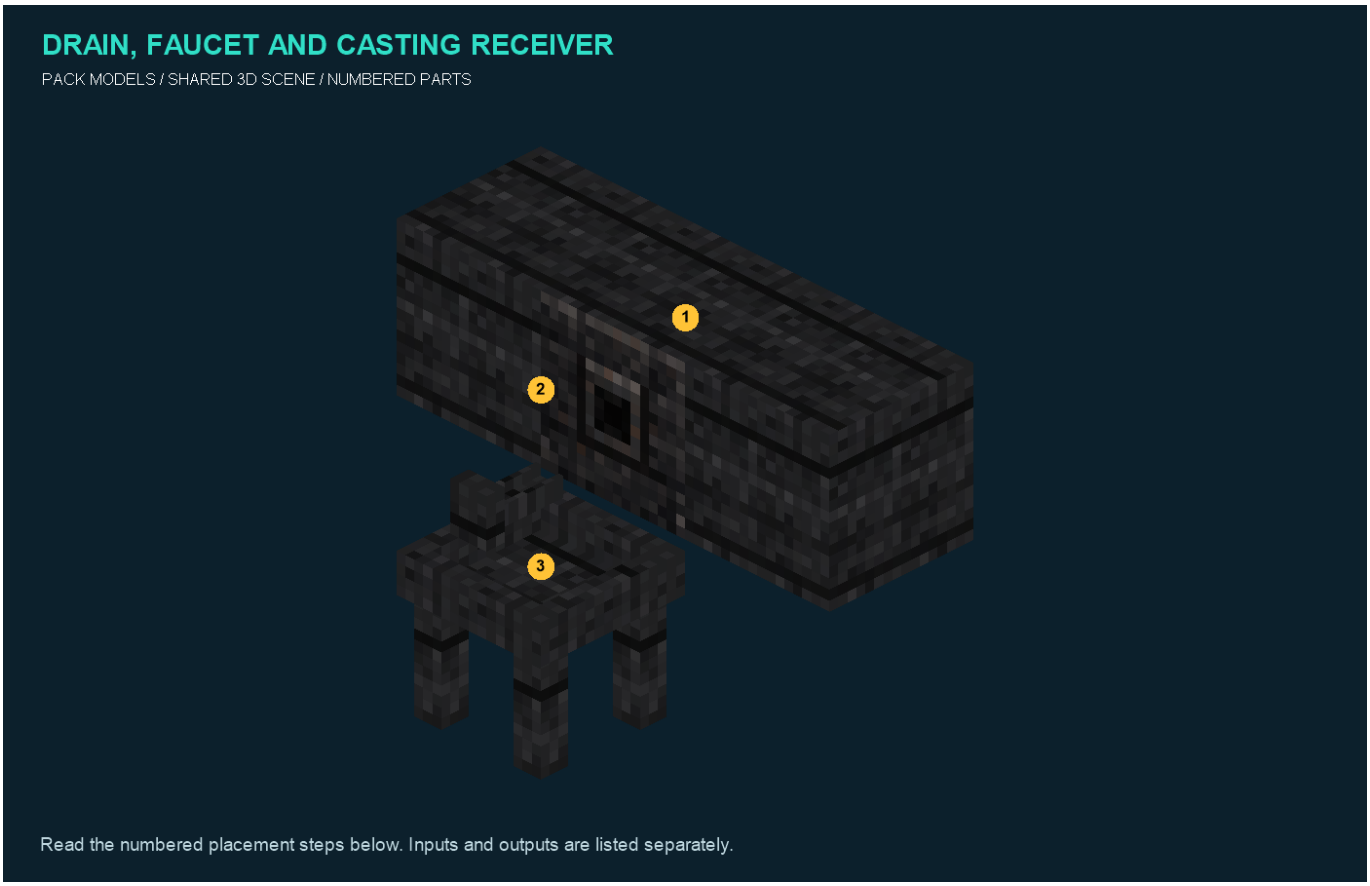
**3.** Active main sources supply the network while charging the bank. A full bank stops drawing charging power.

**4.** When no active primary source is available, the bank supplies its stored energy. An active but undersized source does not automatically become a battery-assisted mode.

***READ MORE:*** *Generators, Watt Pipes and battery backup* | *All build examples*

***CRAFT THE BLOCKS:*** *Battery Bank* | *Watt Pipe* | *Solar Panel*

## Drain, faucet and casting receiver




***INPUT:*** *A valid smeltery + selected molten metal + correct cast/receiver*

***OUTPUT:*** *Cooled ingot, part or metal block*

**1.** Build a valid smeltery first. Put the Drain in its wall and select the intended molten material at the Controller.

**2.** Place a Faucet outside the Drain, with the Casting Table directly below its pour path. A Casting Basin can replace the Table for bulk recipes; it is not a second outlet.

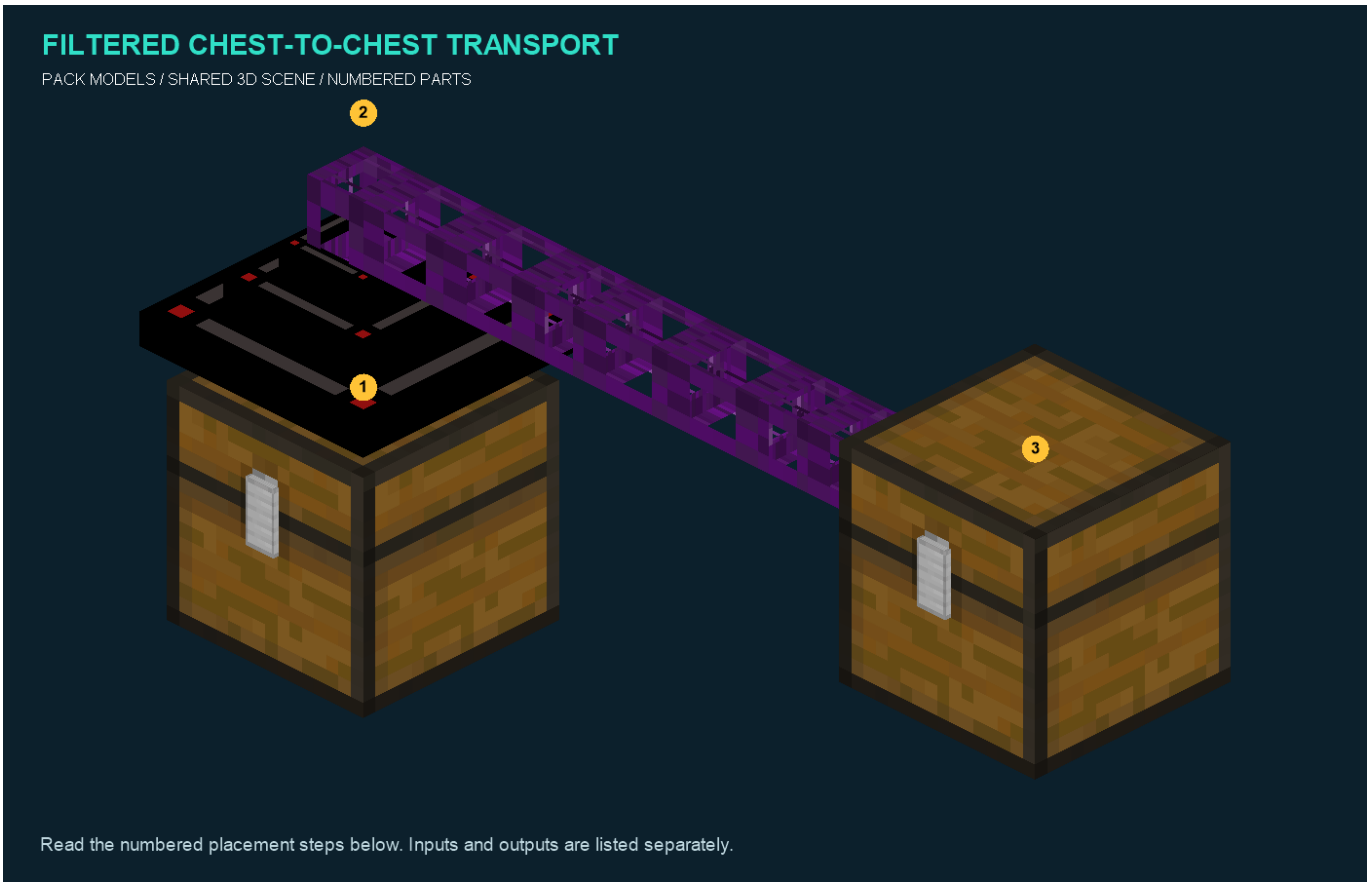
**3.** Use the Table for compatible casts and tool parts. Use the Basin for bulk block casting. Supply enough metal for the selected recipe.

**4.** Wait for the pour and cooling to finish, then collect. Do not replace the receiver while it contains an unfinished pour.

***READ MORE:*** *Alloys, reusable casts and the Tool Forge* | *All build examples*

***CRAFT THE BLOCKS:*** *Smeltery Drain* | *Seared Faucet* | *Casting Table* | *Casting Basin*

## Filtered chest-to-chest transport




***INPUT:*** *Source inventory + listed filter items*

***OUTPUT:*** *Only permitted items reach destinations*

**1.** Place a Transfer Node on top of the source chest; the default intake is down. Confirm the intake side if you rotate or reconfigure it.

**2.** Connect purple Transfer Pipes to a destination inventory. The second chest is the destination in this example.

**3.** Install the circuit board to enable smart routing. Put real filter examples in its persistent filter slots.

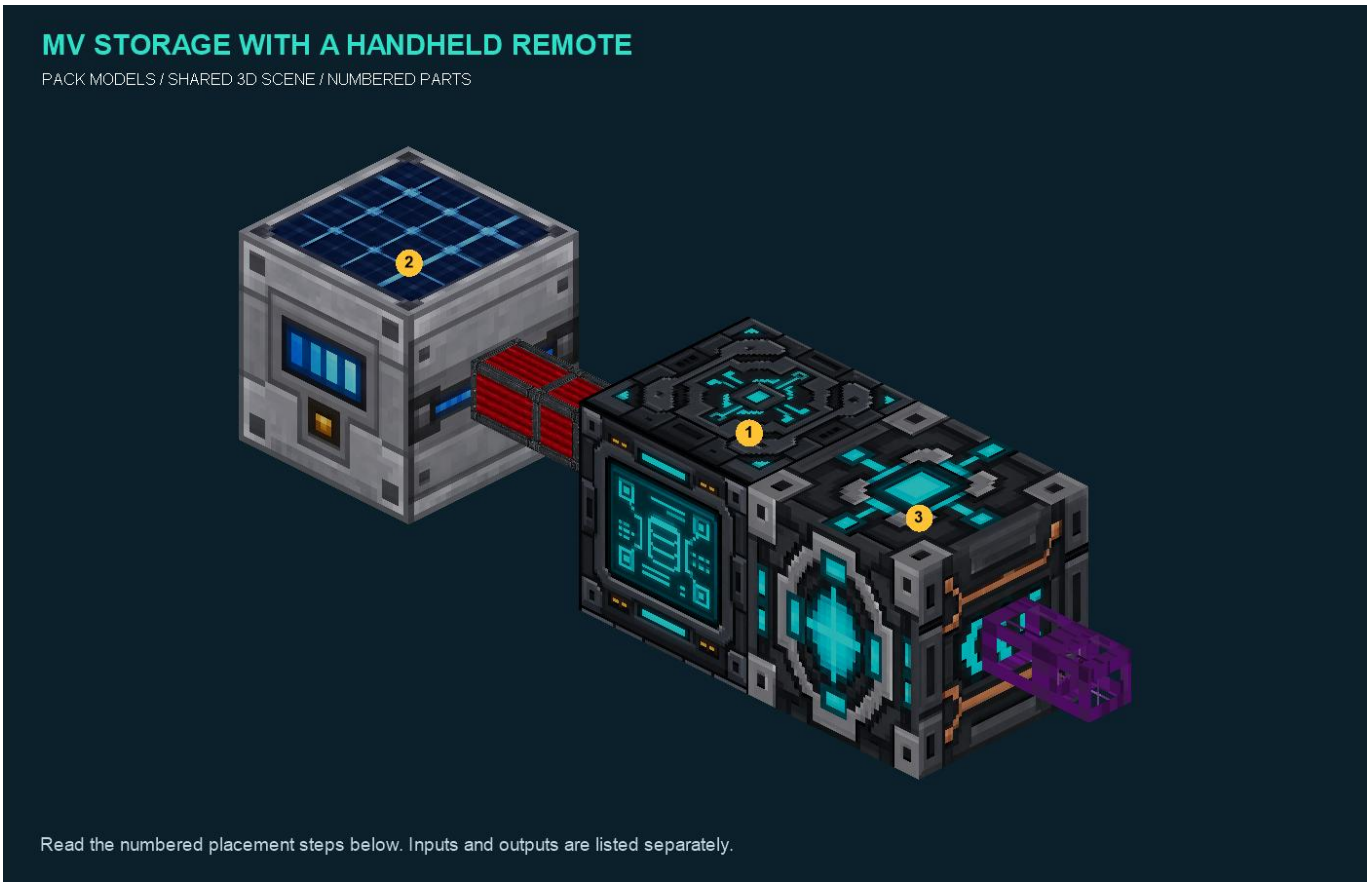
**4.** ALLOW sends only the listed types; BLOCK rejects the listed types. Test one allowed and one blocked item before turning on the full farm. Set destination priorities if several inventories are connected.

***READ MORE:*** *Transfer Nodes, filters, priorities and bulk storage* | *All build examples*

***HUD:*** *Inputs, outputs and controls*

***CRAFT THE BLOCKS:*** *Transfer Node* | *Transfer Pipe* | *MV Link*

## MV Storage with a handheld remote




***INPUT:*** *Powered MV Storage + bound handheld remote*

***OUTPUT:*** *Remote storage and integrated crafting access*

**1.** Build and power MV Storage. The remote opening check requires at least 100 W available at the bound MV system.

**2.** Sneak-use the handheld Wireless MV Remote directly on MV Storage to bind it. Release Sneak, then use the remote to open storage.

**3.** Keep the target loaded in the same dimension. The retired Wireless MV Link access point is not required.

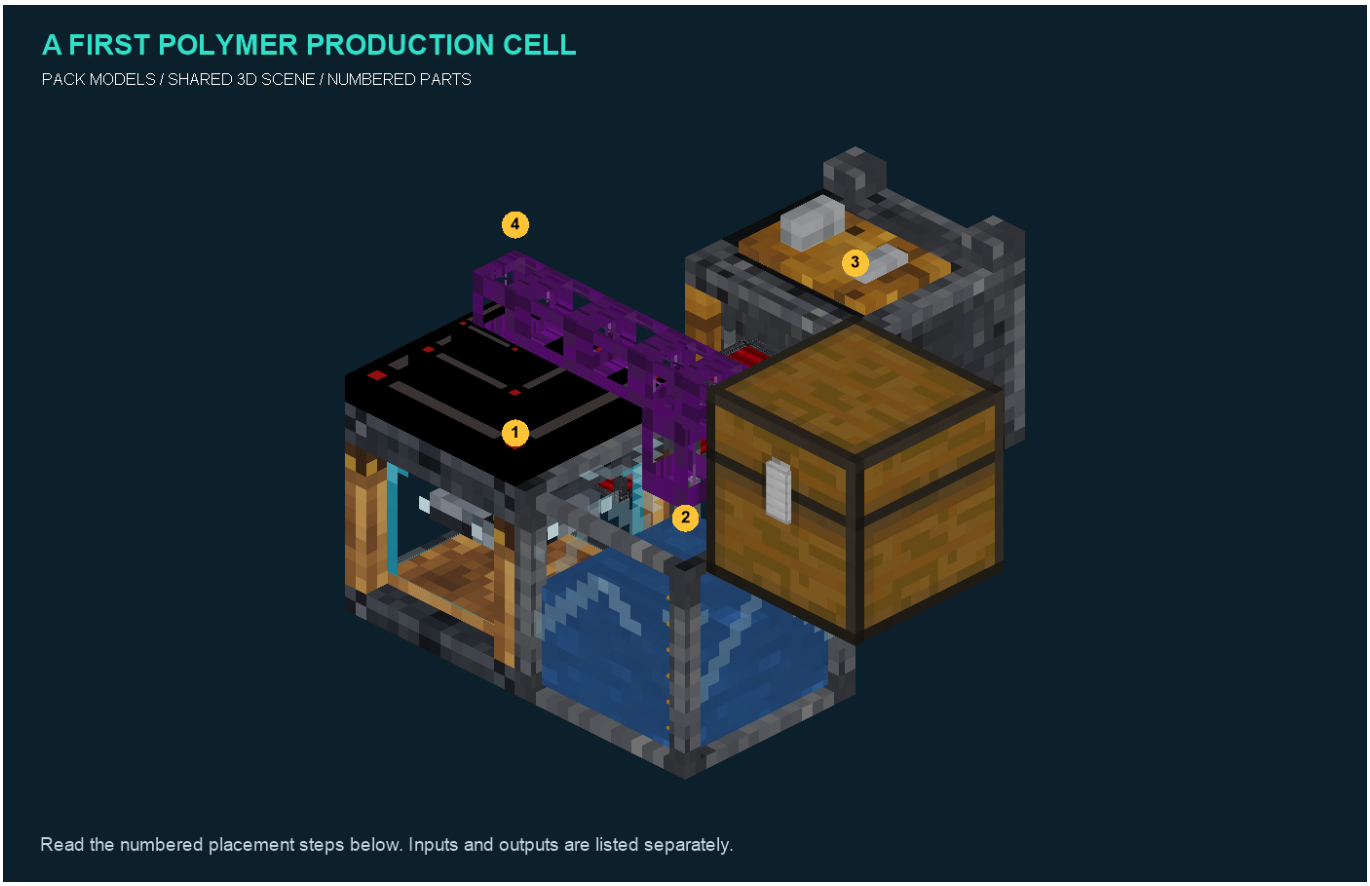
**4.** For automatic pipe insertion/extraction, keep the wired MV Link Unit. Remote access does not replace item-transfer plumbing.

***READ MORE:*** *MV Storage, in-screen crafting and wireless access* | *All build examples*

***HUD:*** *Inputs, outputs and controls*

***CRAFT THE BLOCKS:*** *MV System* | *MV Link*

## A first Polymer production cell




***INPUT:*** *16 biomass + 1 Charcoal + 1,000 mB water + power*

***OUTPUT:*** *4 Polymer per completed batch*

**1.** Place the Pressure Reactor and connect a sufficient power source. Its unmodified working cost is 400 W per working second.

**2.** Supply 16 accepted crop/sapling biomass items, one Charcoal and 1,000 mB water. An adjacent connected water-tank route can refill the machine.

**3.** A base batch needs eight progress steps and makes four Polymer. Speed and efficiency upgrades change work rate and power use.

**4.** Provide output extraction. Use Polymer for circuit boards and the advanced factory chain instead of stockpiling crops without an outlet.

***READ MORE:*** *Processors and the Polymer factory* | *All build examples*

***HUD:*** *Inputs, outputs and controls*

***CRAFT THE BLOCKS:*** *Pressure Reactor* | *Precision Assembler* | *Material Infuser* | *Factory Core*

## Essence infusion without power




***INPUT:*** *A real Infusion Crystal + 4 lower-tier essence*

***OUTPUT:*** *1 next-tier essence per operation*

**1.** Craft the Infusion Crystal at a normal Crafting Table: four Prosperity Shards, four Gold Ingots and a Diamond. The Essence Table does not craft this raw recipe.

**2.** In the Essence Tinkering Table, place the crystal plus four lower-tier essence. No generator is required.

**3.** Process the colored essence tiers in order. Crops produce Inferium Essence; they do not directly replace the table's 4-to-1 tier upgrades.

**4.** Use the linked armor chapter for ingots, full sets and charms. The table's input and output inventories must have room.

***READ MORE:*** *Essence armor: the complete five-tier progression* | *All build examples*

***HUD:*** *Inputs, outputs and controls*

***CRAFT THE BLOCKS:*** *Essence Tinkering Table*

## An Activator aimed at a sapling




***INPUT:*** *A valid target + usable input/tool*

***OUTPUT:*** *Repeated configured tool/use actions*

**1.** Place the target where the Activator is aimed. The drawing only shows relative positions; set its actual working face in the HUD.

**2.** Load Bone Meal or the supported tool into input/tool slots. Choose Place/Use, enable the machine, and check its face and aim.

**3.** It spends tools in first-slot order and stops when the current tool wears out. Keep output storage clear.

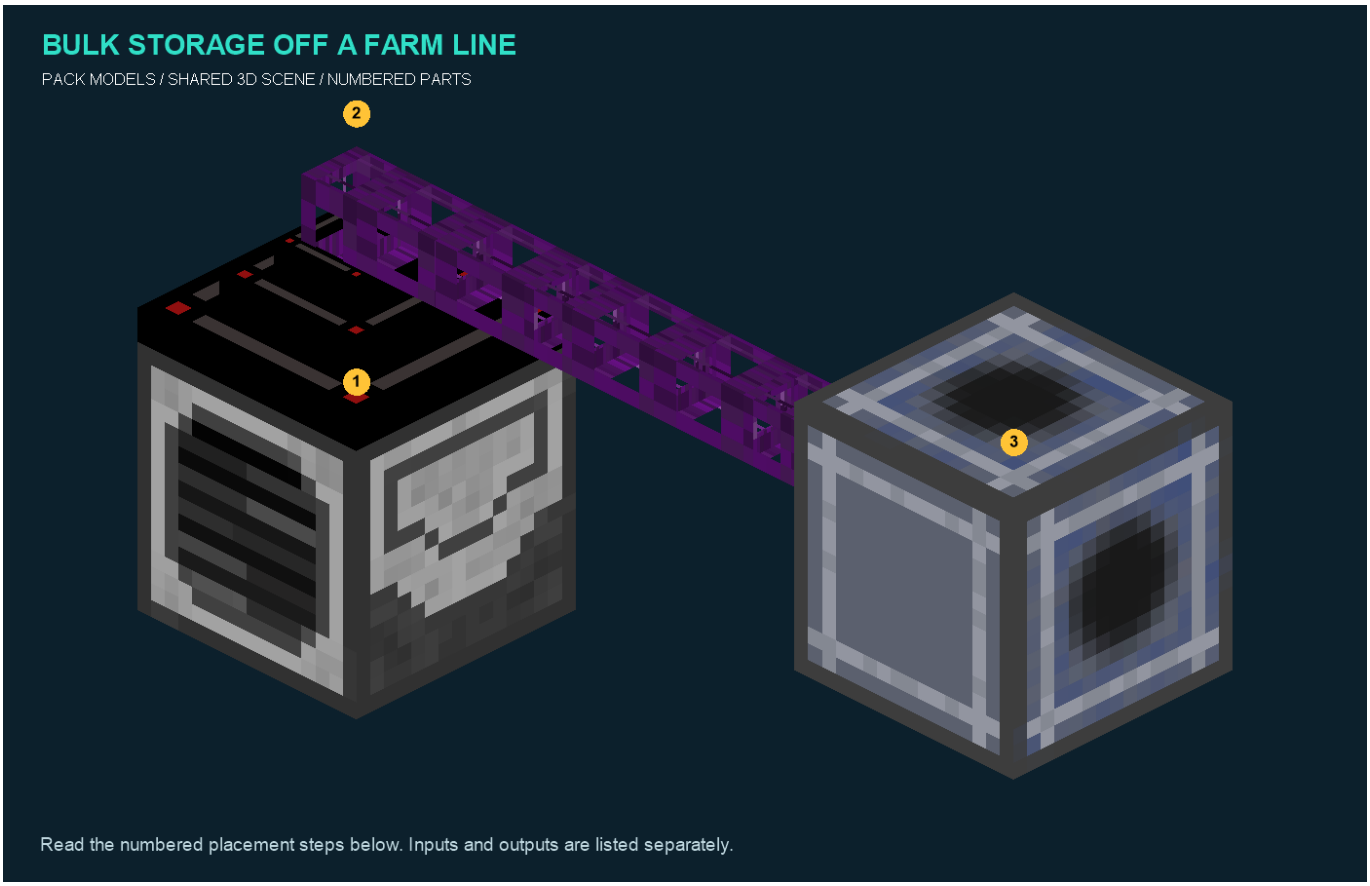
**4.** Use only supported upgrade slots. Speed boards increase action rate, with three active upgrades total; removing them restores the base rate.

***READ MORE:*** *Planters, harvesters, sprinklers and Activators* | *All build examples*

***HUD:*** *Inputs, outputs and controls*

***CRAFT THE BLOCKS:*** *Activator*

## Bulk storage off a farm line




***INPUT:*** *A high-volume single item type*

***OUTPUT:*** *Bulk storage with compatible pipe access*

**1.** Attach an extraction node to the farm's output inventory. Confirm its intake is aimed at the farm.

**2.** Run its Transfer Pipe route to the Black Hole Unit. Reserve the unit for the chosen bulk item type.

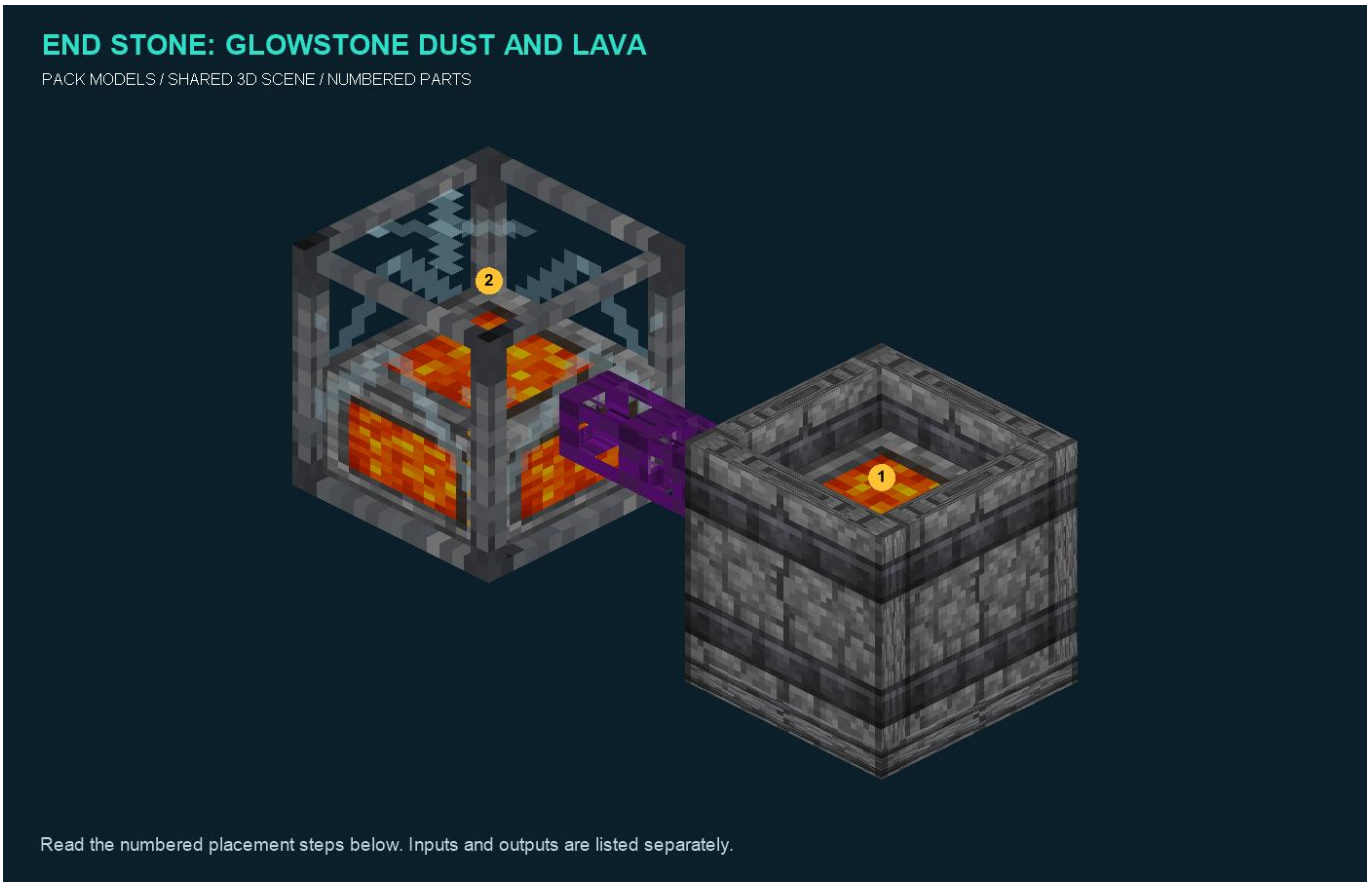
**3.** Use smart filters when other drops share the source inventory. Route secondary drops to other storage destinations.

**4.** The Black Hole Unit supports insertion and extraction through compatible transfer routes. Test both directions with a small batch.

***READ MORE:*** *Transfer Nodes, filters, priorities and bulk storage* | *All build examples*

***CRAFT THE BLOCKS:*** *Black Hole Unit* | *Transfer Node*

## End Stone: Glowstone Dust and lava




***INPUT:*** *1 Glowstone Dust + 100 mB lava*

***OUTPUT:*** *1 End Stone*

**1.** Place Stone Barrel 1. For manual operation, fill it with a Lava Bucket; each End Stone uses only 100 mB of the 1,000 mB bucket.

**2.** Insert 1 Glowstone Dust for each End Stone. Collect the result from the barrel output before it fills. Redstone in the same barrel makes Netherrack instead.

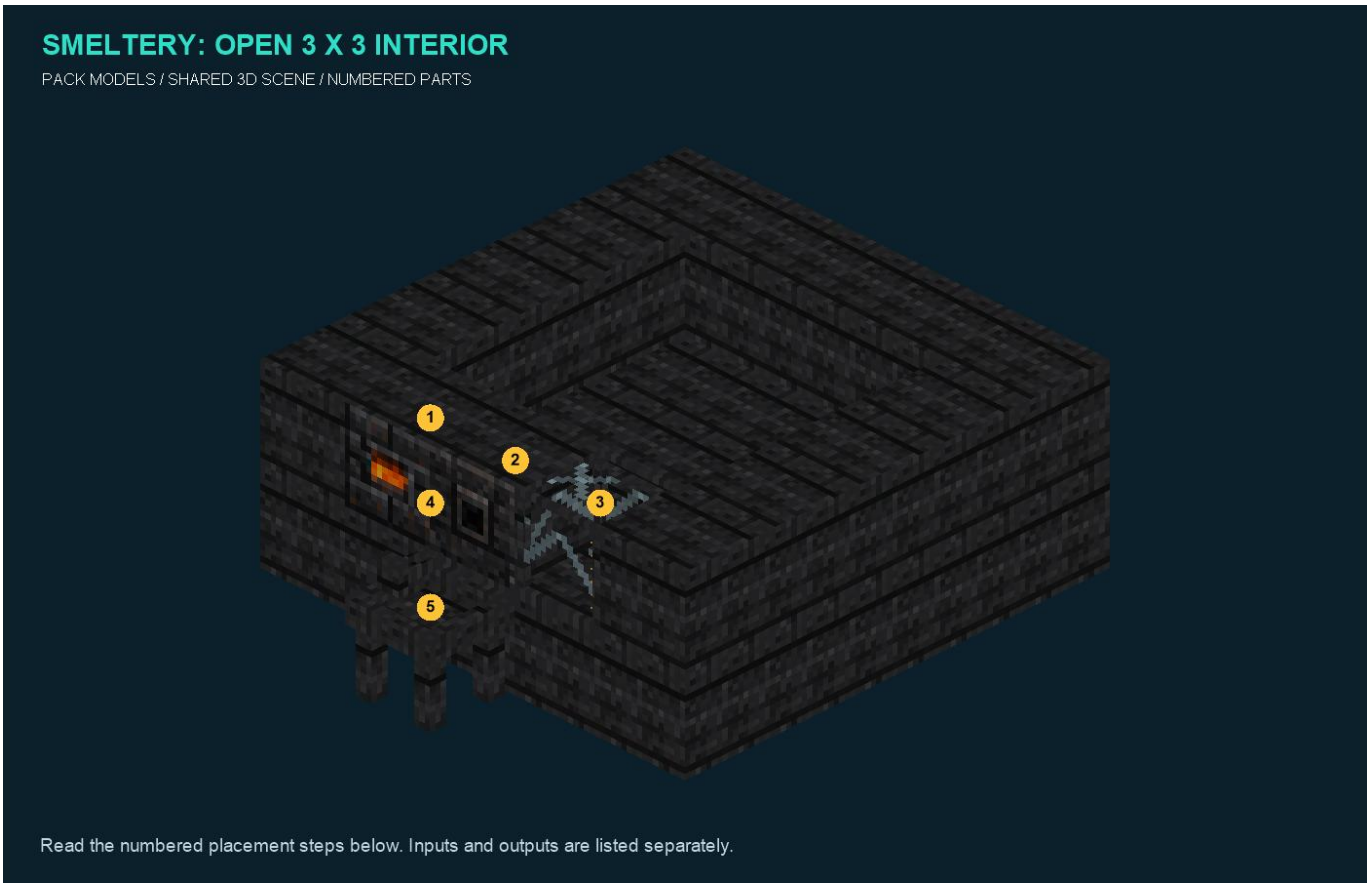
**3.** Optional renewable supply: connect lava Tank 2 through a Transfer Pipe to the barrel. The barrel pulls only the lava needed for its next conversion while output has room. Fill the tank from your lava production first.

**4.** For item automation, use a Transfer Node to supply Glowstone Dust and another correctly facing extraction node to collect End Stone. Keep the input filter on Glowstone Dust. The two-block illustration shows only the barrel and optional lava supply.

***READ MORE:*** *Material recipes: End Stone and other progression blocks* | *All build examples*

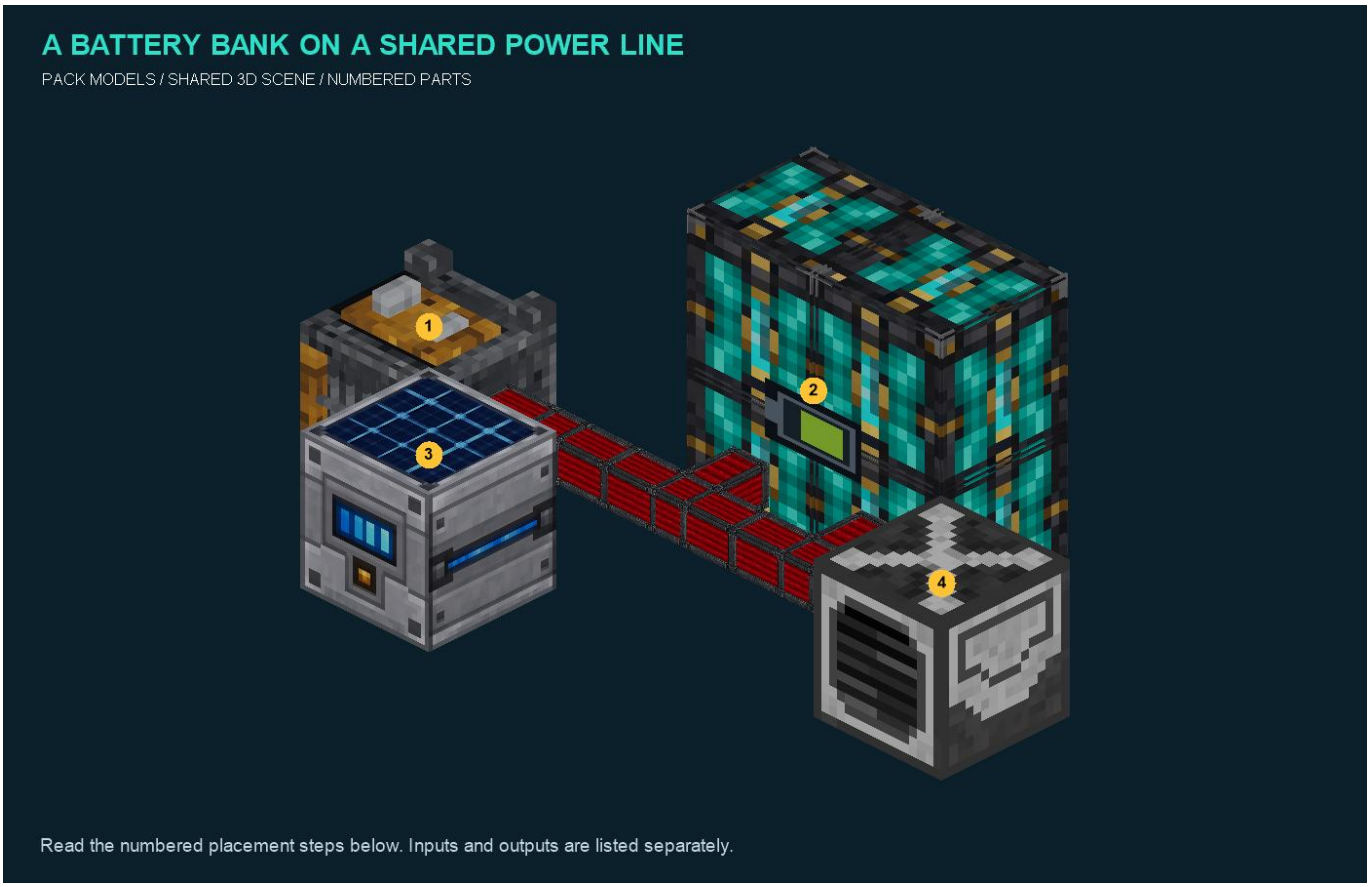
***CRAFT THE BLOCKS:*** *Stone Barrel* | *Tank* | *Transfer Pipe*

# Smeltery construction plan




*Build a 5 × 5 Seared Brick floor and a perimeter wall above it. The four outside corners of the floor and each wall layer are optional. Leave the center 3 × 3 open. Put the Controller, a Drain and a lava-filled Seared Tank in the wall. Place the Faucet outside the Drain and a Casting Table or Casting Basin immediately below the Faucet.*

# Power, fluid and item build plans



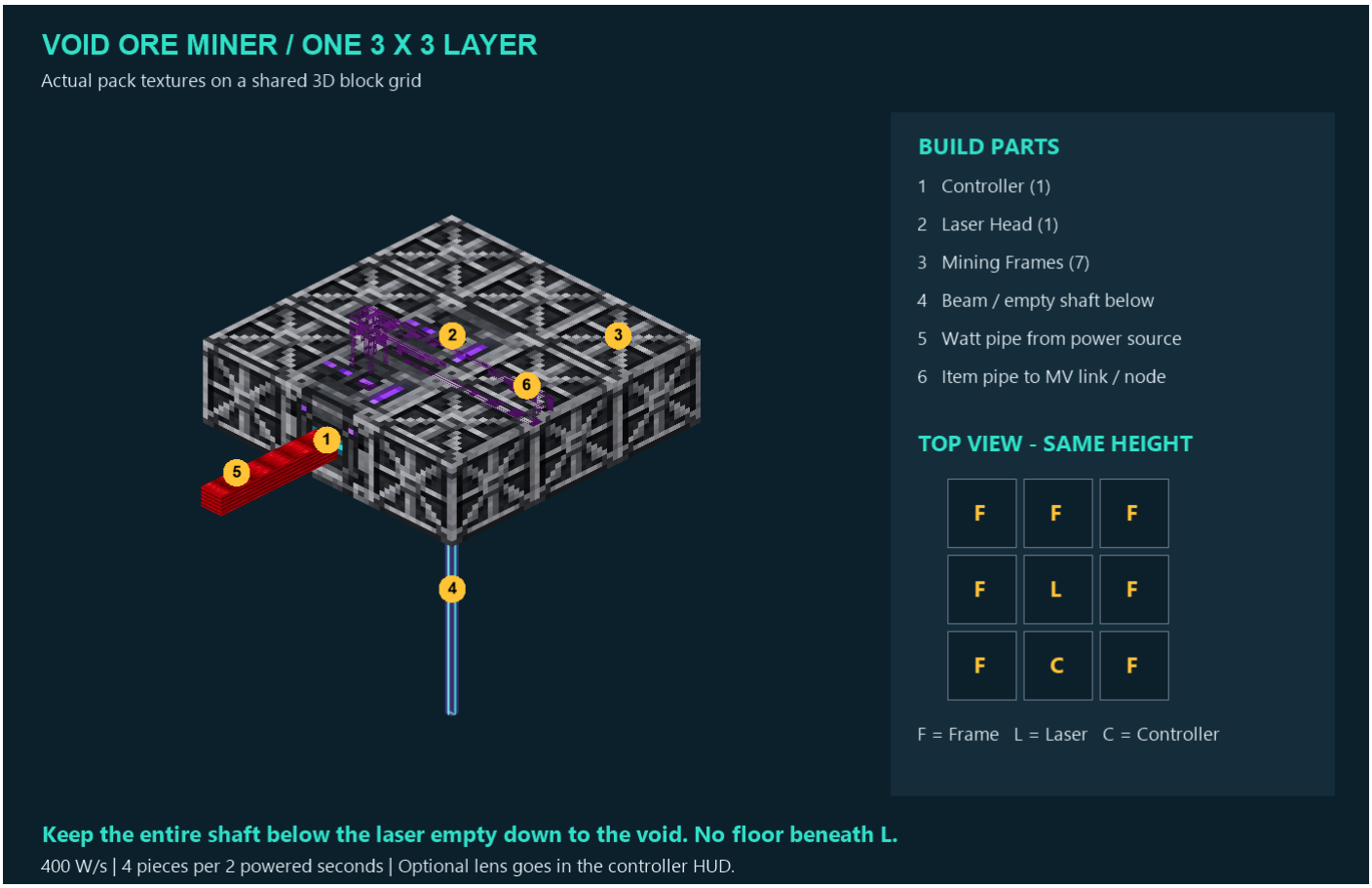

Power: connect sources, batteries and machines to the same Watt Pipe network. Battery backup is a branch of this shared network, not a required pass-through block.

Water: place the water-producing composter above a Moisture Collector. Place the Fluid Pump against the valid water source and connect its tank route.

Items: attach a Transfer Node to the machine output and connect its Transfer Pipe route to an MV Link connected to an MV system. Test a small batch before extending the network.

# Void Ore Miner: build the laser platform

Late-game metal production without gravel or sand input. A complete structure produces four pieces of one raw/crushed metal every two powered seconds. Sieves still provide seeds, essence and specialty progression drops.




| Build part | Placement |
|---|---|
| 7 Mining Frames + 1 Controller | Make the outside ring of a 3 x 3 square. The Controller replaces the middle frame on any one side. |
| 1 Laser Head | Place in the center at the same height as the ring. The whole structure is one layer, nine blocks total. |
| Clear shaft | Leave every block below the center Laser Head empty to the bottom of the Overworld. Build over the island edge; water, lava and passable blocks obstruct it. |
| Connections | Attach Watt pipes to the Controller. Sneak-right-click with a Transfer Node to attach extraction, then connect it to storage. Alternatively use an enabled MV Transfer link through item pipes. |

***NEXT:*** *Parts and lens recipes* | *Power, output and troubleshooting* | *Contents*

# Void miner: parts and Focus Lens recipes

All seven recipes below are shapeless: ingredient positions do not matter. Use a crafting table or register the same ingredients in the Universal Crafter. Recipes are taken from the pack definitions.

| Item / output | Ingredients |
|---|---|
|  Void Mining Frame x4 | 2 Steel Ingot + 4 Seared Brick + 1 Reinforced Frame |
|  Void Laser Head x1 | 2 Crystalline Alloy + 2 Diamond + 1 Conductive Core + 4 Glass |
|  Void Ore Miner Controller x1 | 4 Reinforced Frame + 2 Advanced Circuit + 1 Conductive Core + 1 Diamond + 1 Void Laser Head |
|  Iron Focus Lens x1 | 4 Glass + 4 Iron Ingot + 1 Light Gray Dye |
|  Copper Focus Lens x1 | 4 Glass + 4 Copper Ingot + 1 Orange Dye |
|  Gold Focus Lens x1 | 4 Glass + 4 Gold Ingot + 1 Yellow Dye |
|  Alloy Focus Lens x1 | 4 Glass + 4 Steel Ingot + 1 Purple Dye |

For one complete miner, craft TWO Laser Heads: one is consumed by the Controller recipe and the other is placed in the center. Craft Mining Frames twice for eight frames, then place seven.

***JUMP TO:*** *Build rendering* | *Operating the miner* | *Contents*

# Void miner: power, lenses and output

| Setting | Current behavior |
|---|---|
| Production | 4 pieces of the selected raw/crushed metal per completed 2-second cycle. No input blocks are consumed. |
| Working power | 400 W per working second; 800 W per four-piece batch. Supply sustained generation or a charged battery reserve. |
| Lens slot | The single slot at the left of the Controller HUD is optional. Search Focus Lens to find Iron, Copper, Gold or Alloy. Lenses are reusable; stacking does not increase their effect. |
| Controls | Pause/Resume enables production. Rescan Structure immediately rechecks the ring and shaft. Collect Output moves results into available player inventory space. |
| Output storage | 27 output slots. Connect extraction to MV or another inventory. Normal right-click opens the HUD; sneak-right-click places a Transfer Node against the machine. |

## Focus Lens effects

| Lens | Favored output | Approximate chance per batch |
|---|---|---|
| None | Default mixed metal distribution | See base chances below |
| Iron | Raw Iron | 68.3% instead of 35% |
| Copper | Raw Copper | 57.1% instead of 25% |
| Gold | Raw Gold | 35.3% instead of 12% |
| Alloy | Six custom crushed metals combined | 60.9% instead of 28% |

Lenses multiply matching weights by four and then normalize the table. They do not guarantee a metal, increase total output, speed up the miner or change power consumption. All four pieces in one batch are the same selected metal.

## Base chances and processing

Raw Iron 35%; Raw Copper 25%; Raw Gold 12%; Crushed Tin 7%; Crushed Silver 6%; Crushed Lead 5%; Crushed Nickel 5%; Crushed Aluminum 4%; Crushed Platinum 1%. Process outputs using the existing furnace, pulverizer, smeltery and crafting routes; these are not finished ingots.

## If it stops or the beam is missing

Check the Controller status first: complete the ring, clear the entire shaft, supply 400 W per working second, remove unsupported lens-slot items and extract full output. Waiting conditions preserve progress and resume when fixed. The beam appears only during powered production. The machine does not force chunks to stay loaded.

***JUMP TO:*** *Build rendering* / *Crafting recipes* / *Power and transport builds* / *Contents*

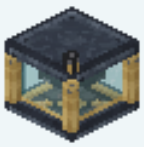
# Recipes made inside production machines

These recipes run inside their named machine, not at a Crafting Table. Insert the ingredients into INPUT, connect Watt Pipes, supply water where listed, and leave output space. Recipe order in the input slots does not matter. The machine automatically processes the first complete supported recipe; its recipe-guide button only changes the displayed instructions.

Rates below are base values without upgrades. W/s is the working power draw; W/job is the total base energy for one successful batch. Machines wait when required ingredients, water, power or output space are missing.

***MACHINE GUIDES:*** *pressure reactor* | *precision assembler* | *material infuser* | *factory core*

# Pressure Reactor — machine recipes



16 supported biomass + Charcoal + Water → Polymer. Biomass may be mixed: crops, supported saplings and Mangrove Propagules. Supply water buckets or a connected water tank.

| Output / batch | Inputs and operating cost |
|---|---|
| **Polymer ×4**  | 16 Crops or saplings + 1 Charcoal<br>Water: 1000 mB consumed per batch<br>8 seconds \| 400 W/s \| 3200 W/job |

***CONTINUE:*** *Production recipes* | *Machine directory* | *Contents*

# Precision Assembler — machine recipes



Insert all listed ingredients in INPUT. Collect finished items from OUTPUT, or extract them through your item network. Use Next recipe guide to browse the machine's recipes.

| Output / batch | Inputs and operating cost |
|---|---|
| **Reinforced Frame ×1**  | 4 Steel Ingot + 2 Polymer + 1 Bronze Ingot<br>Water: 0 mB consumed per batch<br>8 seconds \| 250 W/s \| 2000 W/job |
| **Advanced Circuit ×2**  | 2 Polymer Circuit Board + 2 Electrum Ingot + 1 Quartz + 2 Polymer<br>Water: 0 mB consumed per batch<br>10 seconds \| 500 W/s \| 5000 W/job |
| **Conductive Core ×1**  | 2 Platinum Ingot + 2 Invar Ingot + 1 Redstone Block + 1 Advanced Circuit<br>Water: 0 mB consumed per batch<br>15 seconds \| 1000 W/s \| 15000 W/job |
| **Growth Essence ×4**  | 16 Wheat + 4 Bone Meal + 1 Polymer<br>Water: 0 mB consumed per batch<br>6 seconds \| 250 W/s \| 1500 W/job |

***CONTINUE:*** *Production recipes* | *Machine directory* | *Contents*

# Precision Assembler — machine recipes (continued)



Insert all listed ingredients in INPUT. Collect finished items from OUTPUT, or extract them through your item network. Use Next recipe guide to browse the machine's recipes.

| Output / batch | Inputs and operating cost |
|---|---|
| **Reinforcement Plate ×1**  | 1 Reinforced Frame + 2 Crystalline Alloy<br>Water: 0 mB consumed per batch<br>10 seconds \| 750 W/s \| 7500 W/job |
| **Efficiency Module ×1**  | 2 Advanced Circuit + 1 Crystalline Alloy + 8 Redstone<br>Water: 0 mB consumed per batch<br>10 seconds \| 750 W/s \| 7500 W/job |
| **Fortune Module ×1**  | 2 Advanced Circuit + 2 Crystalline Alloy + 16 Lapis Lazuli + 2 Diamond<br>Water: 0 mB consumed per batch<br>15 seconds \| 1000 W/s \| 15000 W/job |

***CONTINUE:*** *Production recipes* | *Machine directory* | *Contents*

# Material Infuser — machine recipes



Insert all listed ingredients in INPUT. Collect finished items from OUTPUT, or extract them through your item network. Use Next recipe guide to browse the machine's recipes.

| Output / batch | Inputs and operating cost |
|---|---|
| **Crystalline Alloy ×2**  | 2 Diamond + 4 Steel Ingot + 2 Nickel Ingot + 1 Polymer<br>Water: 1000 mB consumed per batch<br>15 seconds \| 1000 W/s \| 15000 W/job |
| **Enriched Essence ×4**  | 8 Growth Essence + 4 Amethyst Shard + 1 Advanced Circuit<br>Water: 1000 mB consumed per batch<br>12 seconds \| 750 W/s \| 9000 W/job |

***CONTINUE:*** *Production recipes* | *Machine directory* | *Contents*

# Factory Core — machine recipes



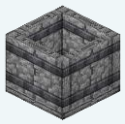
Insert all listed ingredients in INPUT. Collect finished items from OUTPUT, or extract them through your item network. Use Next recipe guide to browse the machine's recipes.

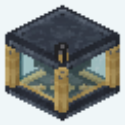
| Output / batch | Inputs and operating cost |
|---|---|
| **Factory Trophy ×1**  | 8 Crystalline Alloy + 4 Conductive Core + 4 Advanced Circuit + 8 Polymer + 4 Platinum Ingot + 4 Pink Slime + 4 Enriched Essence + 1 Netherite Ingot<br>Water: 8000 mB consumed per batch<br>60 seconds \| 2000 W/s \| 120000 W/job |

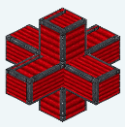
***CONTINUE:*** *Production recipes* | *Machine directory* | *Contents*

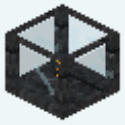
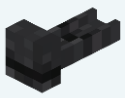
# Machine and utility directory

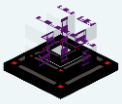
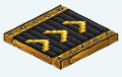
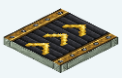
Identify the block, its job and the first setup steps. Use the numbered chapters for complete systems and the crafting appendix for the block recipe.

| | |
|---|---|
|  **Sieve** | Turn dirt, gravel, sand and other supported inputs into resources. Install a mesh, load one input, and click six times. Drops eject into the world; use a hopper or collection system. |
|  **Moisture Collector** | Place beneath the water-producing composter arrangement to collect water over time. A Fluid Pump can extract the collected water. |
|  **Stone Barrel** | Holds lava. Add Redstone Dust for Netherrack, Glowstone Dust for End Stone, or a Water Bucket for Obsidian. Each conversion uses 100 mB of lava. |
|  **Kiln** | Melt one Cobblestone into 1,000 mB of lava. Take the lava with an empty bucket or move it with a Fluid Pump. No power required. |
|  **Magma Furnace** | Smelt supported raw materials into ingots and other products using power. Provide an output route before leaving a batch running. |
|  **Pulverizer** | Grind supported blocks, ores and materials. Processing yields depend on the input; powder is not a full ingot. Check the machine's recipe guide for supported inputs. |
|  **Dehydrator** | Powered drying recipes turn the supported wet/raw inputs into their listed outputs. |
|  **Glacial Chiller** | Powered cooling recipes use their listed inputs and water equivalents. Keep its water supply available. |

| | |
|---|---|
|  <br>**Pressure Reactor** | Turn supported farm products into Polymer with charcoal, water and power. Feed every required ingredient, not just crops. |
|  <br>**Precision Assembler** | Powered advanced-part assembly. Combine the listed ingredients for circuits and advanced factory components. |
|  <br>**Material Infuser** | Powered infusion stage for advanced material upgrades. Supply the exact recipe inputs and extract the finished material. |
|  <br>**Factory Core** | Late-game powered production endpoint. It processes its own recipes; it is not a replacement for every earlier machine. |
|  <br>**Essence Tinkering Table** | No power required. Insert a real Infusion Crystal plus four lower-tier essence, previous-tier equipment plus matching ingots, or eligible gear plus a charm. The raw crystal recipe belongs at a normal crafting table. |
|  <br>**Planter** | Supply supported crop seeds and power. It places crops on valid soil in range. Add a transfer route to replenish seeds. |
|  <br>**Harvester** | Supply power and valid mature crops or trees in range. Collected drops stay in its output slots; attach a Transfer Node to export them. |
|  <br>**Automated Sprinkler** | Place near crops with its required water supply. It supports local growth; it does not harvest or replant. |
|  <br>**Activator** | Load tools or usable items, choose click mode and facing, and enable it. Tools wear one at a time in slot order. Speed upgrades shorten its action interval. |

| | |
|---|---|
|  <br>**Mob Grinder** | Powered mob-killing and drop collection. Provide a safe mob chamber and output extraction. |
|  <br>**Slaughter Factory** | Powered mob-processing route for meat and slime production. Feed mobs into its working area and extract its outputs. |
|  <br>**Mob Duplicator** | Use a supported captured mob and its listed resources to duplicate mobs. Build a secure chamber before activation. |
|  <br>**Combustion Generator** | Fuel-based power source. Active generation feeds the Watt network; it stops when its usable fuel is exhausted. |
|  <br>**Pink Slime Generator** | Late-game power from the pack's pink-slime production chain. Maintain its fuel/input supply. |
|  <br>**Solar Panel** | Solar power source. Connect it to Watt Pipes; generation depends on its operating conditions. |
|  <br>**Advanced Solar Panel** | Higher-tier solar generation. Multiple active sources contribute to the connected power network and battery charging. |
|  <br>**Battery Bank** | Charge from an active source through Watt Pipes. Adjacent batteries combine capacity. Full batteries stop charging; stored energy is reserved for when active primary sources are unavailable. |
|  <br>**Watt Pipe** | Connect power sources, batteries and consumers. It carries network power; it is not a replacement generator. |

| | |
|---|---|
|  **Fluid Pump** | Move supported fluid from an adjacent source into a connected compatible tank route. A water composter on a Moisture Collector is a valid water source. |
|  **Tank** | Connect matching tanks to increase fluid capacity. Connected storage fills from the bottom upward. Check the displayed fluid before mixing supplies. |
|  **Seared Tank** | Smeltery-compatible fluid/fuel tank. Include at least one in the first wall ring and provide lava for smelting. |
|  **Smeltery Controller** | The only Controller in a valid seared smeltery. Open it to load meltable inputs, inspect metals and select the intended pour. |
|  **Smeltery Drain** | Outlet in the smeltery wall. Attach a Faucet and position a casting receiver below the pour. |
|  **Seared Faucet** | Transfer selected molten material from the Drain into the receiver below. An occupied or incompatible receiver stops the pour. |
|  **Casting Table** | Cast ingots, tool parts and reusable casts using the appropriate pattern/cast and molten material. Wait for cooling before extraction. |
|  **Casting Basin** | Receiver for bulk block casting. Place below a Faucet and supply enough of a compatible molten metal. |
|  **Tool Forge** | Assemble modular tools from compatible cast parts; apply supported modifiers through its listed recipes. |

| | |
|---|---|
|  **Transfer Node** | Extract items from the attached inventory or machine and route them through Transfer Pipes. Install a circuit board for persistent smart filters and destination priorities. |
|  **Transfer Pipe** | Item transport connections for Transfer Nodes. These are distinct from fluid and Watt networks. |
|  **Vacuum Hopper** | Collect nearby loose item drops into its inventory. Attach a transfer route to export the collected items. |
|  **MV System** | Shared item storage with search, withdrawal, deposit and an integrated 3 x 3 crafting pattern. Build the network before adding remote access. |
|  **MV Link** | Connect external item transfer to the MV network. Supports item insertion and extraction through compatible transfer routes. |
|  **Universal Crafter** | Choose a supported recipe and supply its ingredients. It supports the pack's registered crafting recipes; browsing an output alone does not provide ingredients. |
|  **Black Hole Unit** | Bulk storage for large quantities of one item type. Compatible transfer routes can insert and extract. |
|  **Conveyor Belt** | Move items/entities along the belt's direction. Build a continuous route and test its orientation before scaling. |
|  **Express Conveyor** | Faster conveyor tier. Use for high-volume surface transport; it is separate from Transfer Pipe routing. |

# Sieve drop tables: every block and mesh

Chance means the probability of getting at least one of that item from one completed sieve input. Repeated rolls for the same item are combined. Several different items can drop together. The figures include the weighted fallback used when every independent roll fails. Rounded to two decimals; very rare results are marked <0.01%.

Amounts show the possible quantity of that item. Dirt guarantees 4 regular pebbles before extra rolls; it is not a 15% pebble drop. These tables document current behavior, without changing it.

## Crooks, infested leaves and hammers

Crooks add a 15% matching-sapling roll and an independent 3% Silkworm roll per natural leaf, in addition to normal leaf loot. Vein mining performs the same rolls separately for each leaf. Better crook materials do not raise these percentages. Infested leaves give 1-3 String when crook-mined through the special handling.

Hammer conversions are guaranteed for supported blocks: Cobblestone -> Gravel -> Sand -> Dust; Netherrack -> Crushed Netherrack. Normally one output per block. Tool tier affects the tool's stats, not a random ore-drop bonus. A supported Tool Forge extra-product modifier can produce an additional output.

## Crushed Netherrack

| Drop / Basic amount | Basic | Iron | Diamond | Netherite |
|---|---|---|---|---|
|  Quartz (1) | 100% | 100% | 100% | 100% |
|  Gold Nugget (0-1) | 20% | 25% | 32% | 40% |
|  Netherite Scrap (0-1) | 1% | 1.25% | 1.60% | 2% |

## Dirt

| Drop / Basic amount | Basic | Iron | Diamond | Netherite |
|---|---|---|---|---|
|  Andesite Pebble (0-2) | 9.84% | 12.25% | 15.59% | 19.36% |
|  Basalt Pebble (0-2) | 9.84% | 12.25% | 15.59% | 19.36% |
|  Blackstone Pebble (0-2) | 9.84% | 12.25% | 15.59% | 19.36% |
|  Diorite Pebble (0-2) | 55% | 67.19% | 83.20% | 100% |
|  Granite Pebble (0-2) | 9.84% | 12.25% | 15.59% | 19.36% |
|  Pebble (4-6) | 100% | 100% | 100% | 100% |
|  Bamboo (0-1) | 5% | 6.25% | 8% | 10% |
|  Beetroot Seeds (0-1) | 5% | 6.25% | 8% | 10% |
|  Sweet Berries (0-1) | 5% | 6.25% | 8% | 10% |
|  Birch Sapling (0-1) | 5% | 6.25% | 8% | 10% |
|  Carrot (0-1) | 5% | 6.25% | 8% | 10% |

| Drop / Basic amount | Basic | Iron | Diamond | Netherite |
|---|---|---|---|---|
| Dark Oak Sapling (0-1) | 5% | 6.25% | 8% | 10% |
| Grass Seeds (0-1) | 7% | 8.75% | 11.20% | 14% |
| Melon Seeds (0-1) | 5% | 6.25% | 8% | 10% |
| Oak Sapling (0-1) | 5% | 6.25% | 8% | 10% |
| Potato (0-1) | 5% | 6.25% | 8% | 10% |
| Pumpkin Seeds (0-1) | 5% | 6.25% | 8% | 10% |
| Spruce Sapling (0-1) | 5% | 6.25% | 8% | 10% |
| Wheat Seeds (0-1) | 7% | 8.75% | 11.20% | 14% |
| Acacia Sapling (0-1) | 4% | 5% | 6.40% | 8% |
| Jungle Sapling (0-1) | 4% | 5% | 6.40% | 8% |
| Cocoa Beans (0-1) | 4% | 5% | 6.40% | 8% |
| Cactus (0-1) | 4% | 5% | 6.40% | 8% |
| Cocoa Seeds (0-1) | 4% | 5% | 6.40% | 8% |
| Sugar Cane Seeds (0-1) | 4% | 5% | 6.40% | 8% |
| Inferium Essence (0-1) | 5% | 6.25% | 8% | 10% |

*Regular pebbles: Basic 4-6 (average 5.25), Iron 4-6 (5.56), Diamond 5-6 (5.8), Netherite exactly 6. Inferium Essence: 5%, 6.25%, 8%, 10% respectively.*

## Gravel

| Drop / Basic amount | Basic | Iron | Diamond | Netherite |
|---|---|---|---|---|
| Coal (0-1) | 16.79% | 18.81% | 22.04% | 26.17% |
| Diamond (0-1) | 2.02% | 2.26% | 2.65% | 3.14% |
| Emerald (0-1) | 2.02% | 2.26% | 2.65% | 3.14% |
| Flint (0-1) | 33.58% | 37.62% | 44.08% | 52.34% |
| Lapis Lazuli (0-1) | 6.72% | 7.52% | 8.82% | 10.47% |
| Aluminum Scrap (0-1) | 6.72% | 7.52% | 8.82% | 10.47% |
| Copper Fragment (0-1) | 8.06% | 9.03% | 10.58% | 12.56% |
| Gold Fragment (0-1) | 4.03% | 4.51% | 5.29% | 6.28% |

| Drop / Basic amount | Basic | Iron | Diamond | Netherite |
|---|---|---|---|---|
| Iron Fragment (0-1) | 26.87% | 30.09% | 35.27% | 41.88% |
| Lead Fragment (0-1) | 4.03% | 4.51% | 5.29% | 6.28% |
| Nickel Fragment (0-1) | 4.03% | 4.51% | 5.29% | 6.28% |
| Platinum Fragment (0-1) | 6.72% | 7.52% | 8.82% | 10.47% |
| Silver Fragment (0-1) | 4.03% | 4.51% | 5.29% | 6.28% |
| Tin Fragment (0-1) | 8.06% | 9.03% | 10.58% | 12.56% |

## Sand

| Drop / Basic amount | Basic | Iron | Diamond | Netherite |
|---|---|---|---|---|
| Crushed Iron Ore (0-1) | 23.14% | 27.11% | 33.19% | 40.60% |
| Crushed Gold Ore (0-1) | 4.63% | 5.42% | 6.64% | 8.12% |
| Crushed Copper Ore (0-1) | 11.57% | 13.55% | 16.59% | 20.30% |
| Crushed Silver Ore (0-1) | 5.78% | 6.78% | 8.30% | 10.15% |
| Crushed Lead Ore (0-1) | 5.78% | 6.78% | 8.30% | 10.15% |
| Crushed Nickel Ore (0-1) | 5.78% | 6.78% | 8.30% | 10.15% |
| Crushed Aluminum Ore (0-1) | 9.25% | 10.84% | 13.28% | 16.24% |
| Crushed Tin Ore (0-1) | 9.25% | 10.84% | 13.28% | 16.24% |
| Crushed Platinum Ore (0-1) | 2.31% | 2.71% | 3.32% | 4.06% |
| Certus Quartz (0-1) | 5.78% | 6.78% | 8.30% | 10.15% |
| Bone Meal (0-1) | 13.88% | 16.27% | 19.91% | 24.36% |
| Redstone (0-1) | 9.25% | 10.84% | 13.28% | 16.24% |
| Glowstone Dust (0-1) | 4.63% | 5.42% | 6.64% | 8.12% |
| Blaze Powder (0-1) | 2.31% | 2.71% | 3.32% | 4.06% |
| Gunpowder (0-1) | 4.63% | 5.42% | 6.64% | 8.12% |
| Cactus (0-1) | 5.78% | 6.78% | 8.30% | 10.15% |
| Cocoa Seeds (0-1) | 4.63% | 5.42% | 6.64% | 8.12% |
| Sugar Cane Seeds (0-1) | 5.78% | 6.78% | 8.30% | 10.15% |

| Drop / Basic amount | Basic | Iron | Diamond | Netherite |
|---|---|---|---|---|
| Cocoa Beans (0-1) | 3.47% | 4.07% | 4.98% | 6.09% |
| Wheat Seeds (0-1) | 3.47% | 4.07% | 4.98% | 6.09% |
| Melon Seeds (0-1) | 3.47% | 4.07% | 4.98% | 6.09% |
| Ancient Spores (0-1) | 5.78% | 6.78% | 8.30% | 10.15% |
| Prosperity Shard (0-1) | 9.25% | 10.84% | 13.28% | 16.24% |
| Cactus Seeds (0-1) | 5.78% | 6.78% | 8.30% | 10.15% |

## Dust

| Drop / Basic amount | Basic | Iron | Diamond | Netherite |
|---|---|---|---|---|
| Blaze Powder (0-1) | 6.21% | 7.11% | 8.52% | 10.28% |
| Bone Meal (0-1) | 24.84% | 28.44% | 34.08% | 41.13% |
| Glowstone Dust (0-1) | 7.76% | 8.89% | 10.65% | 12.85% |
| Gunpowder (0-1) | 8.69% | 9.95% | 11.93% | 14.40% |
| Redstone (0-1) | 15.53% | 17.77% | 21.30% | 25.71% |
| Iron Ore Powder (0-1) | 24.84% | 28.44% | 34.08% | 41.13% |
| Gold Ore Powder (0-1) | 4.97% | 5.69% | 6.82% | 8.23% |
| Copper Ore Powder (0-1) | 12.42% | 14.22% | 17.04% | 20.56% |
| Silver Ore Powder (0-1) | 6.21% | 7.11% | 8.52% | 10.28% |
| Lead Ore Powder (0-1) | 6.21% | 7.11% | 8.52% | 10.28% |
| Nickel Ore Powder (0-1) | 6.21% | 7.11% | 8.52% | 10.28% |
| Aluminum Ore Powder (0-1) | 9.94% | 11.38% | 13.63% | 16.45% |
| Tin Ore Powder (0-1) | 9.94% | 11.38% | 13.63% | 16.45% |
| Platinum Ore Powder (0-1) | 2.48% | 2.84% | 3.41% | 4.11% |

## Soul Sand

| Drop / Basic amount | Basic | Iron | Diamond | Netherite |
|---|---|---|---|---|
| Quartz (1) | 100% | 100% | 100% | 100% |
| Nether Wart (0-1) | 15% | 18.75% | 24% | 30% |

| Drop / Basic amount | Basic | Iron | Diamond | Netherite |
|---|---|---|---|---|
| Ghast Tear (0-1) | 2% | 2.50% | 3.20% | 4% |

## End Stone

| Drop / Basic amount | Basic | Iron | Diamond | Netherite |
|---|---|---|---|---|
| Ender Pearl (0-1) | 23.65% | 24.87% | 27% | 30% |
| Chorus Flower (0-1) | 7.88% | 8.29% | 9% | 10% |
| Quartz (0-1) | 78.84% | 82.90% | 89.99% | 100% |

## Red Sand

| Drop / Basic amount | Basic | Iron | Diamond | Netherite |
|---|---|---|---|---|
| Crushed Iron Ore (0-1) | 23.14% | 27.11% | 33.19% | 40.60% |
| Crushed Gold Ore (0-1) | 4.63% | 5.42% | 6.64% | 8.12% |
| Crushed Copper Ore (0-1) | 11.57% | 13.55% | 16.59% | 20.30% |
| Crushed Silver Ore (0-1) | 5.78% | 6.78% | 8.30% | 10.15% |
| Crushed Lead Ore (0-1) | 5.78% | 6.78% | 8.30% | 10.15% |
| Crushed Nickel Ore (0-1) | 5.78% | 6.78% | 8.30% | 10.15% |
| Crushed Aluminum Ore (0-1) | 9.25% | 10.84% | 13.28% | 16.24% |
| Crushed Tin Ore (0-1) | 9.25% | 10.84% | 13.28% | 16.24% |
| Crushed Platinum Ore (0-1) | 2.31% | 2.71% | 3.32% | 4.06% |
| Certus Quartz (0-1) | 5.78% | 6.78% | 8.30% | 10.15% |
| Bone Meal (0-1) | 13.88% | 16.27% | 19.91% | 24.36% |
| Redstone (0-1) | 9.25% | 10.84% | 13.28% | 16.24% |
| Glowstone Dust (0-1) | 4.63% | 5.42% | 6.64% | 8.12% |
| Blaze Powder (0-1) | 2.31% | 2.71% | 3.32% | 4.06% |
| Gunpowder (0-1) | 4.63% | 5.42% | 6.64% | 8.12% |
| Cactus (0-1) | 5.78% | 6.78% | 8.30% | 10.15% |
| Cocoa Seeds (0-1) | 4.63% | 5.42% | 6.64% | 8.12% |
| Sugar Cane Seeds (0-1) | 5.78% | 6.78% | 8.30% | 10.15% |

| Drop / Basic amount | Basic | Iron | Diamond | Netherite |
|---|---|---|---|---|
| Cocoa Beans (0-1) | 3.47% | 4.07% | 4.98% | 6.09% |
| Wheat Seeds (0-1) | 3.47% | 4.07% | 4.98% | 6.09% |
| Melon Seeds (0-1) | 3.47% | 4.07% | 4.98% | 6.09% |
| Ancient Spores (0-1) | 5.78% | 6.78% | 8.30% | 10.15% |
| Prosperity Shard (0-1) | 9.25% | 10.84% | 13.28% | 16.24% |
| Cactus Seeds (0-1) | 5.78% | 6.78% | 8.30% | 10.15% |

# Ingredient alternatives

Recipe cards name A001-style groups when any of several ingredients can fill one ingredient role. The representative picture is only the first option. Every permitted alternative is listed here.

# Alphabetical item and recipe index

Click a recipe reference to jump to its crafting grid. This index contains only the added blocks with crafting recipes shown in this guide.

| Item / block | Obtain / recipe references |
|---|---|
| Activator | #122 |
| Advanced Solar Panel | #95 |
| Aluminum Block | #97 |
| Automated Sprinkler | #121 |
| Battery Bank | #126 |
| Black Hole Unit | #127 |
| Bronze Block | #130 |
| Casting Basin | #143 |
| Casting Table | #144 |
| Combustion Generator | #152 |
| Conveyor Belt | #153 - #154 |
| Cooked Meat Block | #155 - #156 |
| Dehydrator | #165 |
| Electrum Block | #177 |
| Essence Tinkering Table | #190 |
| Express Conveyor | #191 |
| Factory Core | #192 |
| Fluid Pump | #194 |
| Glacial Chiller | #196 |
| Harvester | #206 |
| Invar Block | #234 |
| Kiln | #255 |
| Lead Block | #261 |
| Magma Furnace | #281 |
| Material Infuser | #283 |
| Mob Duplicator | #285 |
| Mob Grinder | #286 |

| Item / block | Obtain / recipe references |
|---|---|
| Moisture Collector | #287 |
| MV Link | #288 |
| MV System | #289 |
| Nickel Block | #295 |
| Pink Slime Generator | #324 |
| Planter | #325 |
| Platinum Block | #327 |
| Precision Assembler | #346 |
| Pressure Reactor | #347 |
| Pulverizer | #352 |
| Raw Meat Block | #358 |
| Seared Bricks | #366 |
| Seared Faucet | #367 |
| Seared Glass | #368 |
| Seared Tank | #369 |
| Sieve | #370 |
| Silver Block | #373 |
| Slaughter Factory | #392 |
| Smeltery Controller | #393 |
| Smeltery Drain | #394 |
| Solar Panel | #396 |
| Steel Block | #400 |
| Stone Barrel | #412 |
| Tank | #421 |
| Tin Block | #423 |
| Tool Forge | #442 |
| Transfer Node | #443 |
| Transfer Pipe | #444 |
| Universal Crafter | #445 |

| Item / block | Obtain / recipe references |
| --- | --- |
|  Vacuum Hopper | #447 |
|  Watt Pipe | #449 |

# Added-block crafting recipes

Only added machine, storage, utility and building-block crafting recipes appear here. Vanilla, tool, armor and seed crafting grids are omitted. Machine uses are described in the system chapters. Ingredient pictures show the current pack artwork.

## Crafting table / Universal Crafter - 1 - 8 of 60

### #122 Activator x1
Shapeless: any arrangement



1 Piston; 1 Chest; 2 Iron Ingot; 2 Tin Ingot; 1 Redstone
BUILD & USE THIS BLOCK

### #95 Advanced Solar Panel x1
Shapeless: any arrangement



4 Solar Panel; 1 Diamond; 1 Quartz; 1 Silver Ingot; 1 Nickel Ingot; 1 Redstone Block
MACHINE USES

### #97 Aluminum Block x1
Shapeless: any arrangement



9 Aluminum Ingot
MACHINE USES

### #121 Automated Sprinkler x1
Shapeless: any arrangement



1 Clay Bucket; 1 Iron Ingot; 2 Tin Ingot; 1 Transfer Pipe; 1 Redstone
BUILD & USE THIS BLOCK

### #126 Battery Bank x1
Shaped crafting grid



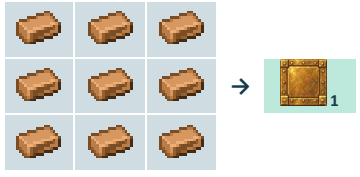
4 Iron Ingot; 2 Copper Ingot; 2 Redstone; 1 Quartz
BUILD & USE THIS BLOCK

### #127 Black Hole Unit x1
Shaped crafting grid



4 Iron Ingot; 4 Redstone; 1 Chest
BUILD & USE THIS BLOCK

### #130 Bronze Block x1
Shapeless: any arrangement



9 Bronze Ingot
MACHINE USES

### #143 Casting Basin x1
Shaped crafting grid



7 Seared Brick
BUILD & USE THIS BLOCK

## Crafting table / Universal Crafter - 9 - 16 of 60

### #144 Casting Table x1
Shaped crafting grid



7 Seared Brick
BUILD & USE THIS BLOCK

### #152 Combustion Generator x1
Shapeless: any arrangement



1 Furnace; 2 Copper Ingot; 1 Nickel Ingot; 1 Aluminum Ingot; 1 Redstone
MACHINE USES

### #153 Conveyor Belt x8
Shapeless: any arrangement



2 Steel Ingot; 2 Bronze Ingot; 2 Aluminum Ingot; 1 Tin Ingot; 2 String
MACHINE USES

### #154 Conveyor Belt x4
Shapeless: any arrangement



2 Bronze Ingot; 1 Tin Ingot; 3 String
MACHINE USES

### #155 Cooked Meat Block x1
Shapeless: any arrangement



1 Raw Meat Block; 1 Charcoal
MACHINE USES

### #156 Cooked Meat Block x1
Shapeless: any arrangement



1 Raw Meat Block; 1 Coal
MACHINE USES

### #165 Dehydrator x1
Shaped crafting grid



4 Silver Ingot; 1 Iron Bars; 2 Iron Ingot; 1 Furnace; 1 Redstone
MACHINE USES

### #177 Electrum Block x1
Shapeless: any arrangement



9 Electrum Ingot
MACHINE USES

## Crafting table / Universal Crafter - 17 - 24 of 60

### #190 Essence Tinkering Table x1
Shaped crafting grid



2 Prosperity Shard; 1 Inferium Essence; 5 Stone; 1 Crafting Table
BUILD & USE THIS BLOCK

### #191 Express Conveyor x4
Shapeless: any arrangement



4 Conveyor Belt; 1 Steel Ingot; 2 Invar Ingot; 2 Electrum Ingot
MACHINE USES

### #192 Factory Core x1
Shapeless: any arrangement



4 Reinforced Frame; 2 Conductive Core; 2 Advanced Circuit; 1 Crystalline Alloy
BUILD & USE THIS BLOCK

### #194 Fluid Pump x1
Shapeless: any arrangement



1 Clay Bucket; 1 Transfer Pipe; 1 Iron Ingot; 2 Tin Ingot
BUILD & USE THIS BLOCK

### #196 Glacial Chiller x1
Shaped crafting grid



4 Tin Ingot; 1 Glass; 2 Nickel Ingot; 1 Furnace; 1 Redstone
MACHINE USES

### #206 Harvester x1
Shapeless: any arrangement



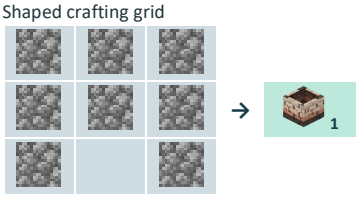
1 Transfer Node; 1 Iron Ingot; 1 Nickel Ingot; 1 Redstone; 1 Iron Hoe
BUILD & USE THIS BLOCK

### #234 Invar Block x1
Shapeless: any arrangement



9 Invar Ingot
MACHINE USES

### #255 Kiln x1
Shaped crafting grid



8 Cobblestone
MACHINE USES

## Crafting table / Universal Crafter - 25 - 32 of 60

### #261 Lead Block x1
Shapeless: any arrangement



9 Lead Ingot
MACHINE USES

### #281 Magma Furnace x1
Shaped crafting grid



4 Iron Ingot; 1 Nickel Ingot; 2 Copper Ingot; 1 Furnace; 1 Tin Ingot
MACHINE USES

### #283 Material Infuser x1
Shapeless: any arrangement



4 Reinforced Frame; 2 Advanced Circuit; 1 Fluid Pump; 2 Invar Ingot
BUILD & USE THIS BLOCK

### #285 Mob Duplicator x1
Shapeless: any arrangement



1 Steel Ingot; 1 Diamond Block; 2 Ender Pearl; 2 Nickel Ingot; 1 Silver Ingot; 1 Safari Net; 1 Chest
MACHINE USES

### #286 Mob Grinder x1
Shapeless: any arrangement



2 Steel Ingot; 2 Invar Ingot; 1 Electrum Ingot; 1 Nickel Ingot; 1 Platinum Ingot; 1 Diamond Sword; 1 Tank
MACHINE USES

### #287 Moisture Collector x1
Shapeless: any arrangement



1 Kiln; 2 Aluminum Ingot; 1 Copper Ingot; 1 Glass; 1 Clay Bucket
BUILD & USE THIS BLOCK

### #288 MV Link x1
Shapeless: any arrangement



1 Transfer Node; 1 Silver Ingot; 1 Redstone
BUILD & USE THIS BLOCK

### #289 MV System x1
Shapeless: any arrangement



1 Netherite Ingot; 1 Diamond Block; 1 Chest; 1 Universal Crafter; 1 Steel Ingot; 1 Silver Ingot; 1 Lead Ingot; 1 Nickel Ingot; 1 Redstone Block
BUILD & USE THIS BLOCK

## Crafting table / Universal Crafter - 33 - 40 of 60

### #295 Nickel Block x1
Shapeless: any arrangement



9 Nickel Ingot
MACHINE USES

### #324 Pink Slime Generator x1
Shapeless: any arrangement



2 Steel Ingot; 2 Invar Ingot; 2 Electrum Ingot; 1 Platinum Ingot; 1 Redstone Block; 1 Combustion Generator
MACHINE USES

### #325 Planter x1
Shapeless: any arrangement



1 Transfer Node; 1 Iron Ingot; 1 Aluminum Ingot; 1 Redstone; 1 Flower Pot
BUILD & USE THIS BLOCK

### #327 Platinum Block x1
Shapeless: any arrangement



9 Platinum Ingot
MACHINE USES

### #346 Precision Assembler x1
Shapeless: any arrangement



4 Steel Ingot; 2 Polymer Circuit Board; 2 Piston; 1 Polymer
BUILD & USE THIS BLOCK

### #347 Pressure Reactor x1
Shapeless: any arrangement



2 Steel Ingot; 2 Invar Ingot; 1 Bronze Ingot; 1 Piston; 1 Fluid Pump; 1 Furnace; 1 Redstone Block
BUILD & USE THIS BLOCK

### #352 Pulverizer x1
Shaped crafting grid



4 Iron Ingot; 1 Piston; 2 Copper Ingot; 1 Furnace; 1 Redstone
MACHINE USES

### #358 Raw Meat Block x1
Shaped crafting grid



9 Pink Essence Bucket
Returns nine empty Buckets.
MACHINE USES

## Crafting table / Universal Crafter - 41 - 48 of 60

### #366 Seared Bricks x1
Shaped crafting grid



4 Seared Brick
MACHINE USES

### #367 Seared Faucet x1
Shaped crafting grid



3 Seared Brick
BUILD & USE THIS BLOCK

### #368 Seared Glass x2
Shaped crafting grid



4 Seared Brick; 1 Glass
MACHINE USES

### #369 Seared Tank x1
Shaped crafting grid



8 Seared Brick; 1 Glass
MACHINE USES

### #370 Sieve x1
Shapeless: any arrangement



4 Oak Planks; 4 String
BUILD & USE THIS BLOCK

### #373 Silver Block x1
Shapeless: any arrangement



9 Silver Ingot
MACHINE USES

### #392 Slaughter Factory x1
Shapeless: any arrangement



1 Steel Ingot; 1 Diamond Sword; 1 Hopper; 1 Chest; 2 Nickel Ingot; 1 Lead Ingot; 1 Redstone; 1 Iron Block
MACHINE USES

### #393 Smeltery Controller x1
Shaped crafting grid



8 Seared Brick; 1 Furnace
MACHINE USES

## Crafting table / Universal Crafter - 49 - 56 of 60

### #394 Smeltery Drain x1
Shaped crafting grid



6 Seared Brick
BUILD & USE THIS BLOCK

### #396 Solar Panel x1
Shapeless: any arrangement



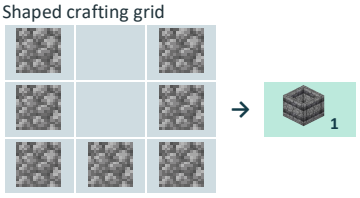
3 Glass; 2 Lapis Lazuli; 1 Quartz; 1 Silver Ingot; 1 Aluminum Ingot; 1 Redstone
BUILD & USE THIS BLOCK

### #400 Steel Block x1
Shapeless: any arrangement



9 Steel Ingot
MACHINE USES

### #412 Stone Barrel x1
Shaped crafting grid



7 Cobblestone
BUILD & USE THIS BLOCK

### #421 Tank x1
Shapeless: any arrangement



2 Cobblestone; 2 Glass; 2 Lead Ingot
BUILD & USE THIS BLOCK

### #423 Tin Block x1
Shapeless: any arrangement



9 Tin Ingot
MACHINE USES

### #442 Tool Forge x1
Shaped crafting grid



3 Iron Block; 5 Seared Bricks; 1 Crafting Table
MACHINE USES

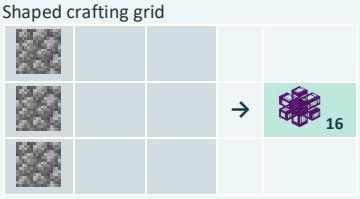
### #443 Transfer Node x1
Shapeless: any arrangement



1 Transfer Pipe; 1 Chest; 1 Iron Ingot
BUILD & USE THIS BLOCK

## Crafting table / Universal Crafter - 57 - 60 of 60

### #444 Transfer Pipe x16
Shaped crafting grid




3 Cobblestone
BUILD & USE THIS BLOCK

### #445 Universal Crafter x1
Shaped crafting grid




4 Aluminum Ingot; 1 Silver Ingot; 1 Nickel Ingot; 1 Crafting Table; 1 MV Link; 1 Redstone Block
MACHINE USES

### #447 Vacuum Hopper x1
Shapeless: any arrangement




1 Hopper; 1 Chest; 1 Ender Pearl; 1 Aluminum Ingot; 1 Transfer Node
BUILD & USE THIS BLOCK

### #449 Watt Pipe x16
Shapeless: any arrangement




2 Cobblestone; 1 Redstone
BUILD & USE THIS BLOCK

# Starting tree to Supremium: costs and difficulty

Supremium is a late-game farming project. Getting the first seeds is the early hurdle; producing tens of thousands of Inferium is the long part. There is no reliable real-time completion estimate because farm size, growth assistance, loading and collection determine production. The numbers below are material calculations for this pack, not SkyFactory Java defaults.

| Stage | What to build or collect |
|---|---|
| 1. Starting tree | Harvest wood and saplings. Compost eight accepted ingredients per dirt. Establish renewable trees before spending your last saplings. |
| 2. Early resources | Make the sieve and hammer chain. Sieve Dirt for Inferium and Wheat Seeds; crush Cobblestone to Gravel to Sand. Sieve Sand for Prosperity Shards. Basic mesh rolls: Inferium 5% per Dirt, Prosperity 8% per Sand. |
| 3. Start essence farming | Craft an Essence Crafting Seed (Wheat Seed + four Prosperity Shards), then the first Inferium seed (crafting seed + four Inferium in a cross). Plant on valid soil and harvest mature crops; the seed remains planted. |
| 4. Infuse and expand | Craft the normal Infusion Crystal at a Crafting Table (four shards, four gold, one diamond). Build the unpowered Essence Tinkering Table. Infuse four lower essence into one higher essence. Expand crops and collection; upgrade seeds for greater Inferium yield. |
| 5. Armor progression | Make Prosperity Ingots, then Inferium Ingots. Craft the initial four Inferium armor pieces. Upgrade every piece through Prudentium, Intermedium, Superium and Supremium using four next-tier ingots per piece in the Essence Table. |
| 6. Supremium | Wear all four pieces for the set bonus. Native flight still requires Education features and Cheats. Tools and optional charms are separate expenses; they are not prerequisites for the armor set. |

## Exact material budget for one armor set

| Armor reached | Ingots used at this step | Cumulative Inferium equivalent |
|---|---|---|
| Inferium | 24 Inferium Ingots | 96 |
| Prudentium | 16 Prudentium Ingots | 416 |
| Intermedium | 16 Intermedium Ingots | 1,760 |
| Superium | 16 Superium Ingots | 7,200 |
| Supremium | 16 Supremium Ingots | 29,024 |

Final armor-only total: 88 Iron Ingots, 352 Prosperity Shards and 29,024 Inferium-equivalent essence. This includes the lower-tier ingots consumed inside higher-tier ingots. It excludes all seeds, extra tools, crystals, the workbench and farm machinery. Do not count the ingots already used in lower armor as reusable ingredients.

## Make the grind manageable

Essence values: 1 Prudentium = 4 Inferium; 1 Intermedium = 16; 1 Superium = 64; 1 Supremium = 256. Ingots also consume the previous ingot, so a Supremium Ingot costs 1 Prosperity Ingot plus 4 essence at EACH of the five tiers: 1,364 Inferium equivalent.

Direct basic-mesh sieving would average 580,480 Dirt for the armor-only essence budget. With Netherite Mesh (10% Inferium roll), the average is 290,240 Dirt. These are expected values, not guarantees, and exclude the other resources. Use sieving to start the farm rather than to finish the set.

| Farm example | Output per complete harvest | Armor-only harvest rounds |
|---|---|---|
| 64 tier-1 plants | 64 Inferium | 454 rounds |
| 64 tier-5 plants | 320 Inferium | 91 rounds |

These examples assume every planted crop matures and is collected once per round. They exclude the essence and shards needed to establish or upgrade the farm, and do not imply a fixed growth time. Higher-tier seeds still produce Inferium, not their matching higher essence.

The armor-only infusion chain uses 9,344 crystal charges: 7,168 Prudentium, 1,728 Intermedium, 384 Superium and 64 Supremium outputs. Budget ten fresh normal crystals at 1,000 uses each, or make a Master Crystal. The Master costs an extra four Supremium Essence (1,024 Inferium equivalent), four Prosperity Ingots and a normal crystal; it then has unlimited uses. Farm upgrades add further infusion work.

Recommended route: get the first seed, expand the renewable farm, add growth assistance and harvesting/collection, then upgrade seeds and infuse stockpiles. A larger reliable farm helps more than manually sieving toward 29,024 essence. Check water, power where required, output space and loaded chunks before investing in speed upgrades.

***RECIPES:*** *All essence recipes* | *Essence workstation setup* | *Crop farm example*

# Essence armor and tools: every recipe

109 recipes from the pack: 14 equipment types at each of five tiers (70 equipment recipes), plus essence conversions, ingots, crystals, crop seeds, charms and watering cans. Crafting cards show exact grids where shaped. PROCESS cards show ingredient quantities only: insert them in any Essence Tinkering Table input slots, unpause, then wait for output. The table requires no power.

## Inferium equipment

Inferium Axe | Inferium Boots | Inferium Chestplate | Inferium Crook | Inferium Hammer | Inferium Helmet | Inferium Hoe | Inferium Leggings | Inferium Paxel | Inferium Pickaxe | Inferium Scythe | Inferium Shovel | Inferium Sickle | Inferium Sword

## Prudentium equipment

Prudentium Axe | Prudentium Boots | Prudentium Chestplate | Prudentium Crook | Prudentium Hammer | Prudentium Helmet | Prudentium Hoe | Prudentium Leggings | Prudentium Paxel | Prudentium Pickaxe | Prudentium Scythe | Prudentium Shovel | Prudentium Sickle | Prudentium Sword

## Intermedium equipment

Intermedium Axe | Intermedium Boots | Intermedium Chestplate | Intermedium Crook | Intermedium Hammer | Intermedium Helmet | Intermedium Hoe | Intermedium Leggings | Intermedium Paxel | Intermedium Pickaxe | Intermedium Scythe | Intermedium Shovel | Intermedium Sickle | Intermedium Sword

## Superium equipment

Superium Axe | Superium Boots | Superium Chestplate | Superium Crook | Superium Hammer | Superium Helmet | Superium Hoe | Superium Leggings | Superium Paxel | Superium Pickaxe | Superium Scythe | Superium Shovel | Superium Sickle | Superium Sword

## Supremium equipment

Supremium Axe | Supremium Boots | Supremium Chestplate | Supremium Crook | Supremium Hammer | Supremium Helmet | Supremium Hoe | Supremium Leggings | Supremium Paxel | Supremium Pickaxe | Supremium Scythe | Supremium Shovel | Supremium Sickle | Supremium Sword

***SUPPORT RECIPES:*** *Blank Essence Charm* / *Efficiency Essence Charm* / *Essence Crafting Seed* / *Fire Resistance Essence Charm* / *Fortune Essence Charm* / *Infusion Crystal* / *Master Infusion Crystal* / *Mining Aoe Essence Charm* / *Night Vision Essence Charm* / *No Fall Essence Charm* / *Prosperity Ingot* / *Resistance Essence Charm* / *Silk Touch Essence Charm* / *Speed Essence Charm* / *Strength Essence Charm* / *Water Breathing Essence Charm* / *Inferium Essence* / *Inferium Essence Seeds* / *Inferium Ingot* / *Inferium Tier2 Essence Seeds* / *Inferium Tier3 Essence Seeds* / *Inferium Tier4 Essence Seeds* / *Inferium Tier5 Essence Seeds* / *Inferium Watering Can* / *Prudentium Essence* / *Prudentium Essence* / *Prudentium Ingot* / *Prudentium Watering Can* / *Intermedium Essence* / *Intermedium Essence* / *Intermedium Ingot* / *Intermedium Watering Can* / *Superium Essence* / *Superium Essence* / *Superium Ingot* / *Superium Watering Can* / *Supremium Essence* / *Supremium Ingot* / *Supremium Watering Can*

## Essence recipes 1 - 6

### Blank Essence Charm x1
Crafting Table / Universal Crafter
Shapeless - any arrangement



1 Prosperity Ingot + 4 Supremium Essence
Essence recipe index

### Efficiency Essence Charm x1
Crafting Table / Universal Crafter
Shapeless - any arrangement



1 Blank Essence Charm + 2 Supremium Essence + 2 Redstone Block
Essence recipe index

### Essence Crafting Seed x1
Crafting Table / Universal Crafter
Shaped - copy the grid



4 Prosperity Shard + 1 Wheat Seeds
Essence recipe index

### Fire Resistance Essence Charm x1
Crafting Table / Universal Crafter
Shapeless - any arrangement



1 Blank Essence Charm + 2 Supremium Essence + 2 Magma Cream
Essence recipe index

### Fortune Essence Charm x1
Crafting Table / Universal Crafter
Shapeless - any arrangement



1 Blank Essence Charm + 2 Supremium Essence + 2 Diamond + 4 Lapis Lazuli
Essence recipe index

### Infusion Crystal x1
Crafting Table / Universal Crafter
Shaped - copy the grid



4 Prosperity Shard + 4 Gold Ingot + 1 Diamond
Essence recipe index

## Essence recipes 7 - 12

### Master Infusion Crystal x1
Crafting Table / Universal Crafter
Shaped - copy the grid



4 Supremium Essence + 4 Prosperity Ingot + 1 Infusion Crystal
Essence recipe index

### Mining Aoe Essence Charm x1
Crafting Table / Universal Crafter
Shapeless - any arrangement



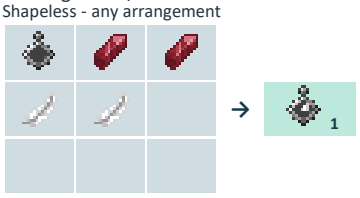
1 Blank Essence Charm + 2 Supremium Essence + 2 Piston
Essence recipe index

### Night Vision Essence Charm x1
Crafting Table / Universal Crafter
Shapeless - any arrangement



1 Blank Essence Charm + 2 Supremium Essence + 2 Golden Carrot
Essence recipe index

### No Fall Essence Charm x1
Crafting Table / Universal Crafter
Shapeless - any arrangement



1 Blank Essence Charm + 2 Supremium Essence + 2 Feather
Essence recipe index

### Prosperity Ingot x1
Crafting Table / Universal Crafter
Shapeless - any arrangement



1 Iron Ingot + 4 Prosperity Shard
Essence recipe index

### Resistance Essence Charm x1
Crafting Table / Universal Crafter
Shapeless - any arrangement



1 Blank Essence Charm + 2 Supremium Essence + 4 Obsidian
Essence recipe index

## Essence recipes 13 - 18

### Silk Touch Essence Charm x1
Crafting Table / Universal Crafter
Shapeless - any arrangement



1 Blank Essence Charm + 2 Supremium Essence + 4 String
Essence recipe index

### Speed Essence Charm x1
Crafting Table / Universal Crafter
Shapeless - any arrangement



1 Blank Essence Charm + 2 Supremium Essence + 4 Sugar
Essence recipe index

### Strength Essence Charm x1
Crafting Table / Universal Crafter
Shapeless - any arrangement



1 Blank Essence Charm + 2 Supremium Essence + 2 Blaze Powder
Essence recipe index

### Water Breathing Essence Charm x1
Crafting Table / Universal Crafter
Shapeless - any arrangement



1 Blank Essence Charm + 2 Supremium Essence + 2 Pufferfish
Essence recipe index

### Inferium Axe x1
Crafting Table / Universal Crafter
Shaped - copy the grid



3 Inferium Ingot + 2 Stick
Essence recipe index

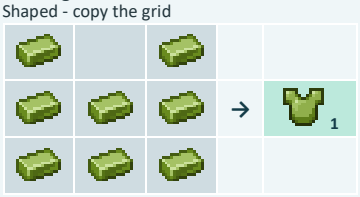
### Inferium Boots x1
Crafting Table / Universal Crafter
Shaped - copy the grid



4 Inferium Ingot
Essence recipe index

## Essence recipes 19 - 24

### Inferium Chestplate x1
Crafting Table / Universal Crafter
Shaped - copy the grid



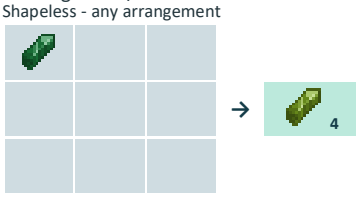
8 Inferium Ingot
Essence recipe index

### Inferium Crook x1
Crafting Table / Universal Crafter
Shaped - copy the grid



3 Inferium Ingot + 2 Stick
Essence recipe index

### Inferium Essence x4
Crafting Table / Universal Crafter
Shapeless - any arrangement



1 Prudentium Essence
Essence recipe index

### Inferium Essence Seeds x1
Crafting Table / Universal Crafter
Shaped - copy the grid



4 Inferium Essence + 1 Essence Crafting Seed
Essence recipe index

### Inferium Hammer x1
Crafting Table / Universal Crafter
Shaped - copy the grid



2 Inferium Ingot + 2 Stick
Essence recipe index

### Inferium Helmet x1
Crafting Table / Universal Crafter
Shaped - copy the grid



5 Inferium Ingot
Essence recipe index

## Essence recipes 25 - 30

### Inferium Hoe x1
Crafting Table / Universal Crafter
Shaped - copy the grid



2 Inferium Ingot + 2 Stick
Essence recipe index

### Inferium Ingot x1
Crafting Table / Universal Crafter
Shapeless - any arrangement



1 Prosperity Ingot + 4 Inferium Essence
Essence recipe index

### Inferium Leggings x1
Crafting Table / Universal Crafter
Shaped - copy the grid



7 Inferium Ingot
Essence recipe index

### Inferium Paxel x1
Crafting Table / Universal Crafter
Shapeless - any arrangement



1 Inferium Pickaxe + 1 Inferium Axe + 1 Inferium Shovel + 1 Prosperity Ingot
Essence recipe index

### Inferium Pickaxe x1
Crafting Table / Universal Crafter
Shaped - copy the grid



3 Inferium Ingot + 2 Stick
Essence recipe index

### Inferium Scythe x1
Crafting Table / Universal Crafter
Shaped - copy the grid



4 Inferium Ingot + 2 Stick
Essence recipe index

## Essence recipes 31 - 36

### Inferium Shovel x1
Crafting Table / Universal Crafter
Shaped - copy the grid



1 Inferium Ingot + 2 Stick
Essence recipe index

### Inferium Sickle x1
Crafting Table / Universal Crafter
Shaped - copy the grid



3 Inferium Ingot + 2 Stick
Essence recipe index

### Inferium Sword x1
Crafting Table / Universal Crafter
Shaped - copy the grid



2 Inferium Ingot + 1 Stick
Essence recipe index

### Inferium Tier2 Essence Seeds x1
Crafting Table / Universal Crafter
Shaped - copy the grid



4 Prudentium Essence + 1 Inferium Essence Seeds
Essence recipe index

### Inferium Tier3 Essence Seeds x1
Crafting Table / Universal Crafter
Shaped - copy the grid



4 Intermedium Essence + 1 Inferium Tier2 Essence Seeds
Essence recipe index

### Inferium Tier4 Essence Seeds x1
Crafting Table / Universal Crafter
Shaped - copy the grid



4 Superium Essence + 1 Inferium Tier3 Essence Seeds
Essence recipe index

## Essence recipes 37 - 42

### Inferium Tier5 Essence Seeds x1
Crafting Table / Universal Crafter
Shaped - copy the grid



4 Supremium Essence + 1 Inferium Tier4 Essence Seeds
Essence recipe index

### Inferium Watering Can x1
Crafting Table / Universal Crafter
Shaped - copy the grid



5 Inferium Ingot + 1 Bucket
Essence recipe index

### Prudentium Axe x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Inferium Axe + 4 Prudentium Ingot

**Output** **1 Prudentium Axe**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

### Prudentium Boots x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Inferium Boots + 4 Prudentium Ingot

**Output** **1 Prudentium Boots**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

### Prudentium Chestplate x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Inferium Chestplate + 4 Prudentium Ingot

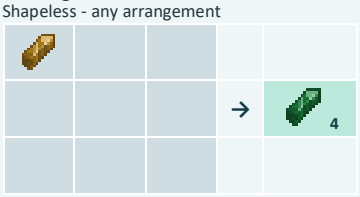
**Output** **1 Prudentium Chestplate**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

### Prudentium Crook x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Inferium Crook + 4 Prudentium Ingot

**Output** **1 Prudentium Crook**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

## Essence recipes 43 - 48

### Prudentium Essence x4
Crafting Table / Universal Crafter
Shapeless - any arrangement



1 Intermedium Essence
Essence recipe index

### Prudentium Essence x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
4 Inferium Essence + 1 Infusion Crystal

**Output** **1 Prudentium Essence**
4s per batch, up to 64 output. Crystal retained; one use per output, 1000 uses per normal crystal.
Essence recipe index

### Prudentium Hammer x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Inferium Hammer + 4 Prudentium Ingot

**Output** **1 Prudentium Hammer**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

### Prudentium Helmet x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Inferium Helmet + 4 Prudentium Ingot

**Output** **1 Prudentium Helmet**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

### Prudentium Hoe x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Inferium Hoe + 4 Prudentium Ingot

**Output** **1 Prudentium Hoe**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

### Prudentium Ingot x1
Crafting Table / Universal Crafter
Shapeless - any arrangement



1 Inferium Ingot + 4 Prudentium Essence
Essence recipe index

## Essence recipes 49 - 54

**Prudentium Leggings x1**
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Inferium Leggings + 4 Prudentium Ingot
**Output**  **1 Prudentium Leggings**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

**Prudentium Paxel x1**
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Inferium Paxel + 4 Prudentium Ingot
**Output**  **1 Prudentium Paxel**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

**Prudentium Pickaxe x1**
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Inferium Pickaxe + 4 Prudentium Ingot
**Output**  **1 Prudentium Pickaxe**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

**Prudentium Scythe x1**
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Inferium Scythe + 4 Prudentium Ingot
**Output**  **1 Prudentium Scythe**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

**Prudentium Shovel x1**
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Inferium Shovel + 4 Prudentium Ingot
**Output**  **1 Prudentium Shovel**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

**Prudentium Sickle x1**
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Inferium Sickle + 4 Prudentium Ingot
**Output**  **1 Prudentium Sickle**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

## Essence recipes 55 - 60

**Prudentium Sword x1**
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Inferium Sword + 4 Prudentium Ingot

**Output** **1 Prudentium Sword**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

**Prudentium Watering Can x1**
Crafting Table / Universal Crafter
Shaped - copy the grid



5 Prudentium Ingot + 1 Inferium Watering Can
Essence recipe index

**Intermedium Axe x1**
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Prudentium Axe + 4 Intermedium Ingot

**Output** **1 Intermedium Axe**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

**Intermedium Boots x1**
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Prudentium Boots + 4 Intermedium Ingot

**Output** **1 Intermedium Boots**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

**Intermedium Chestplate x1**
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Prudentium Chestplate + 4 Intermedium Ingot

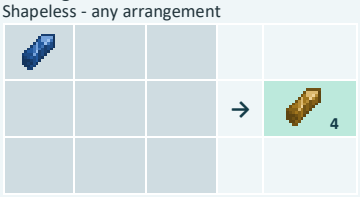
**Output** **1 Intermedium Chestplate**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

**Intermedium Crook x1**
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Prudentium Crook + 4 Intermedium Ingot

**Output** **1 Intermedium Crook**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

## Essence recipes 61 - 66

### Intermedium Essence x4
Crafting Table / Universal Crafter
Shapeless - any arrangement



1 Superium Essence
Essence recipe index

### Intermedium Essence x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
4 Prudentium Essence + 1 Infusion Crystal

**Output** **1 Intermedium Essence**
4s per batch, up to 64 output. Crystal retained; one use per output, 1000 uses per normal crystal.
Essence recipe index

### Intermedium Hammer x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Prudentium Hammer + 4 Intermedium Ingot

**Output** **1 Intermedium Hammer**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

### Intermedium Helmet x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Prudentium Helmet + 4 Intermedium Ingot

**Output** **1 Intermedium Helmet**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

### Intermedium Hoe x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Prudentium Hoe + 4 Intermedium Ingot

**Output** **1 Intermedium Hoe**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

### Intermedium Ingot x1
Crafting Table / Universal Crafter
Shapeless - any arrangement



1 Prudentium Ingot + 4 Intermedium Essence
Essence recipe index

## Essence recipes 67 - 72

**Intermedium Leggings x1**
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Prudentium Leggings + 4 Intermedium Ingot
**Output**  **1 Intermedium Leggings**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

**Intermedium Paxel x1**
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Prudentium Paxel + 4 Intermedium Ingot
**Output**  **1 Intermedium Paxel**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

**Intermedium Pickaxe x1**
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Prudentium Pickaxe + 4 Intermedium Ingot
**Output**  **1 Intermedium Pickaxe**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

**Intermedium Scythe x1**
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Prudentium Scythe + 4 Intermedium Ingot
**Output**  **1 Intermedium Scythe**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

**Intermedium Shovel x1**
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Prudentium Shovel + 4 Intermedium Ingot
**Output**  **1 Intermedium Shovel**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

**Intermedium Sickle x1**
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Prudentium Sickle + 4 Intermedium Ingot
**Output**  **1 Intermedium Sickle**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

## Essence recipes 73 - 78

### Intermedium Sword x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Prudentium Sword + 4 Intermedium Ingot

**Output** **1 Intermedium Sword**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

### Intermedium Watering Can x1
Crafting Table / Universal Crafter
Shaped - copy the grid




5 Intermedium Ingot + 1 Prudentium Watering Can
Essence recipe index

### Superium Axe x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Intermedium Axe + 4 Superium Ingot

**Output** **1 Superium Axe**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

### Superium Boots x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Intermedium Boots + 4 Superium Ingot

**Output** **1 Superium Boots**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

### Superium Chestplate x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Intermedium Chestplate + 4 Superium Ingot

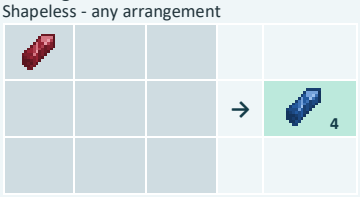
**Output** **1 Superium Chestplate**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

### Superium Crook x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Intermedium Crook + 4 Superium Ingot

**Output** **1 Superium Crook**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

## Essence recipes 79 - 84

### Superium Essence x4
Crafting Table / Universal Crafter
Shapeless - any arrangement



1 Supremium Essence
Essence recipe index

### Superium Essence x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
4 Intermedium Essence + 1 Infusion Crystal

**Output** **1 Superium Essence**
4s per batch, up to 64 output. Crystal retained; one use per output, 1000 uses per normal crystal.
Essence recipe index

### Superium Hammer x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Intermedium Hammer + 4 Superium Ingot

**Output** **1 Superium Hammer**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

### Superium Helmet x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Intermedium Helmet + 4 Superium Ingot

**Output** **1 Superium Helmet**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

### Superium Hoe x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Intermedium Hoe + 4 Superium Ingot

**Output** **1 Superium Hoe**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

### Superium Ingot x1
Crafting Table / Universal Crafter
Shapeless - any arrangement



1 Intermedium Ingot + 4 Superium Essence
Essence recipe index

## Essence recipes 85 - 90

**Superium Leggings x1**
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Intermedium Leggings + 4 Superium Ingot
**Output**  **1 Superium Leggings**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

**Superium Paxel x1**
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Intermedium Paxel + 4 Superium Ingot
**Output**  **1 Superium Paxel**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

**Superium Pickaxe x1**
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Intermedium Pickaxe + 4 Superium Ingot
**Output**  **1 Superium Pickaxe**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

**Superium Scythe x1**
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Intermedium Scythe + 4 Superium Ingot
**Output**  **1 Superium Scythe**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

**Superium Shovel x1**
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Intermedium Shovel + 4 Superium Ingot
**Output**  **1 Superium Shovel**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

**Superium Sickle x1**
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Intermedium Sickle + 4 Superium Ingot
**Output**  **1 Superium Sickle**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

## Essence recipes 91 - 96

**Superium Sword x1**
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Intermedium Sword + 4 Superium Ingot

**Output** **1 Superium Sword**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

**Superium Watering Can x1**
Crafting Table / Universal Crafter
Shaped - copy the grid




5 Superium Ingot + 1 Intermedium Watering Can
Essence recipe index

**Supremium Axe x1**
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Superium Axe + 4 Supremium Ingot

**Output** **1 Supremium Axe**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

**Supremium Boots x1**
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Superium Boots + 4 Supremium Ingot

**Output** **1 Supremium Boots**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

**Supremium Chestplate x1**
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Superium Chestplate + 4 Supremium Ingot

**Output** **1 Supremium Chestplate**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

**Supremium Crook x1**
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Superium Crook + 4 Supremium Ingot

**Output** **1 Supremium Crook**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

## Essence recipes 97 - 102

### Supremium Essence x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
4 Superium Essence + 1 Infusion Crystal

**Output** **1 Supremium Essence**
4s per batch, up to 64 output. Crystal retained; one use per output, 1000 uses per normal crystal.
Essence recipe index

### Supremium Hammer x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Superium Hammer + 4 Supremium Ingot

**Output** **1 Supremium Hammer**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

### Supremium Helmet x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Superium Helmet + 4 Supremium Ingot

**Output** **1 Supremium Helmet**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

### Supremium Hoe x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Superium Hoe + 4 Supremium Ingot

**Output** **1 Supremium Hoe**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

### Supremium Ingot x1
Crafting Table / Universal Crafter
Shapeless - any arrangement




1 Superium Ingot + 4 Supremium Essence
Essence recipe index

### Supremium Leggings x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Superium Leggings + 4 Supremium Ingot

**Output** **1 Supremium Leggings**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

## Essence recipes 103 - 108

### Supremium Paxel x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Superium Paxel + 4 Supremium Ingot

**Output**  **1 Supremium Paxel**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

### Supremium Pickaxe x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Superium Pickaxe + 4 Supremium Ingot

**Output**  **1 Supremium Pickaxe**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

### Supremium Scythe x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Superium Scythe + 4 Supremium Ingot

**Output**  **1 Supremium Scythe**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

### Supremium Shovel x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Superium Shovel + 4 Supremium Ingot

**Output**  **1 Supremium Shovel**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

### Supremium Sickle x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Superium Sickle + 4 Supremium Ingot

**Output**  **1 Supremium Sickle**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

### Supremium Sword x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Superium Sword + 4 Supremium Ingot

**Output**  **1 Supremium Sword**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

## Essence recipes 109 - 109

**Supremium Watering Can x1**
Crafting Table / Universal Crafter
Shaped - copy the grid




5 Supremium Ingot + 1 Superium Watering Can
Essence recipe index

## Charm installation and removal

Finished charm crafting recipes appear above. Install one charm by combining it with a compatible Supremium item in the Essence Tinkering Table. Gear plus Flint removes and returns the charm. Armor flight is the full Supremium set bonus; it does not require a charm.

| Charm | Compatible Supremium targets |
|---|---|
| Speed Essence Charm | Supremium Boots, Supremium Leggings |
| Night Vision Essence Charm | Supremium Helmet |
| Water Breathing Essence Charm | Supremium Helmet |
| Fire Resistance Essence Charm | Supremium Chestplate, Supremium Boots |
| Resistance Essence Charm | Supremium Chestplate |
| Strength Essence Charm | Supremium Sword |
| Fortune Essence Charm | Supremium Pickaxe, Supremium Paxel |
| Silk Touch Essence Charm | Supremium Pickaxe, Supremium Paxel |
| Efficiency Essence Charm | Supremium Pickaxe, Supremium Axe, Supremium Shovel, Supremium Hoe, Supremium Hammer, Supremium Crook, Supremium Paxel, Supremium Sickle, Supremium Scythe |
| Mining Aoe Essence Charm | Supremium Pickaxe, Supremium Axe, Supremium Shovel, Supremium Paxel, Supremium Hammer |
| No Fall Essence Charm | Supremium Boots |

***BACK TO:*** *Essence recipe index* | *Tree-to-Supremium budget*

# About this field guide

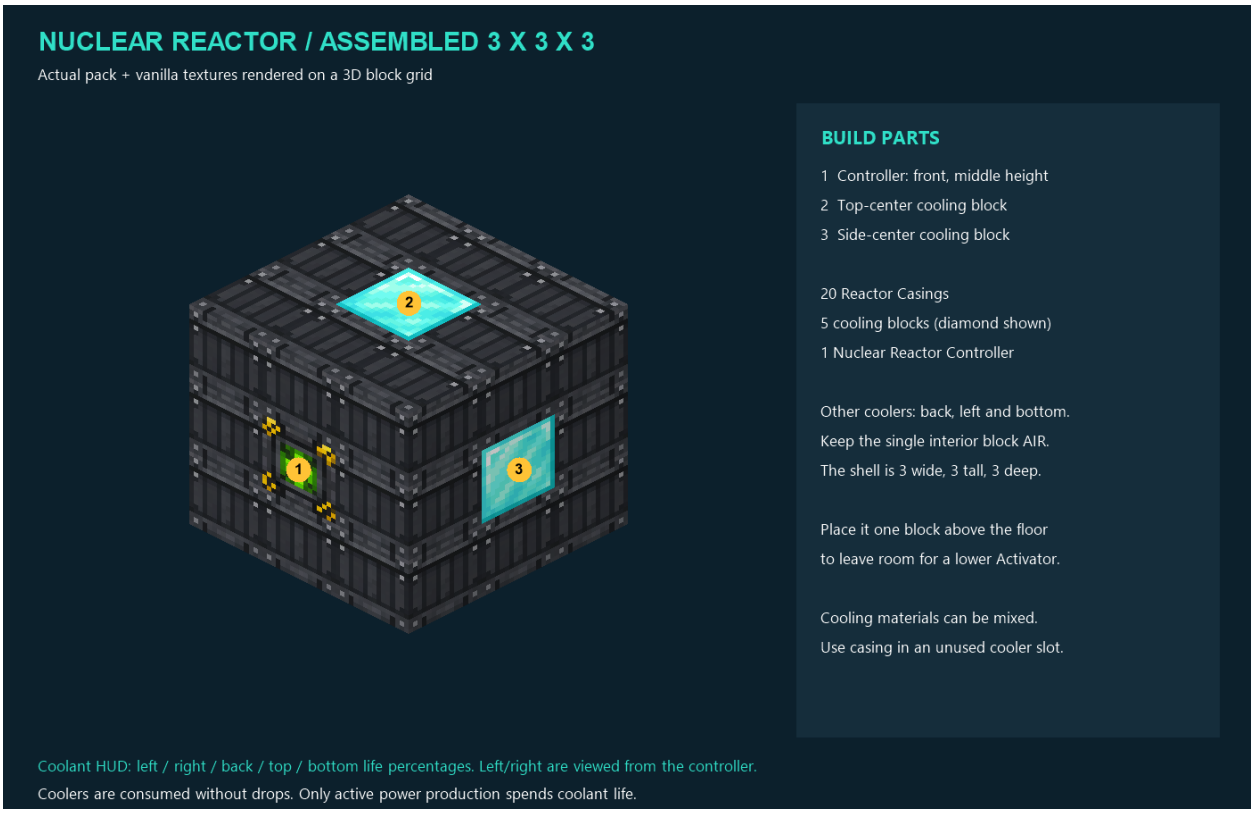
Build 0.2.41. Void miner, beam construction, Focus Lenses and production rates are updated for 0.2.41. Existing guide chapters are retained. This edition includes a step-by-step survival start and 60 added-block crafting recipes. Machine instructions and sieve probabilities come from the pack data. The separate essence appendix includes 109 essence recipes, covering all 70 armor/tool recipes across five tiers. Unrelated vanilla recipes are omitted.

Production-machine recipes now include every Precision Assembler, Material Infuser and Factory Core output plus Pressure Reactor Polymer, with exact ingredients, water, processing time and power. Crafting grids show the finished item and quantity. The Automated Sprinkler illustrations use the 0.2.41 texture. Archived illustrations still appear in the handbook.

Block views use the current block material mappings rather than archived inventory previews. Placement grids and HUD examples are labeled reference schematics, not screenshots. Non-cube items may use their flat icon. Textures belong to the pack creators and Mojang/Microsoft.

No gameplay recipes, drop rates or machine behavior were changed to produce this PDF.

# Nuclear and wireless power




Build a hollow 3 x 3 x 3 shell with the controller in the center of a front wall. Keep the middle block air. Place coolers at the five other face centers. The casing accepts Watt Pipes and wireless transmitters.

One fuel rod supplies 1,200,000 W. Insert rods in the upper input row and extract spent rods from the lower output row. Copper, iron, gold, diamond and netherite coolers increase output in that order. Coolers wear during generation and are destroyed on overheating. Use inward-facing Activators to replace them.

To pair wireless power, place a transmitter next to the source or connect it with Watt Pipes. Note its assigned name. Place another relay at the machine, use Find, and select the transmitter. Keep both ends loaded in the same dimension.

***BACK*** *Contents*

# Flying ships

Build a 3 x 3 deck of Oak Planks and place a Ship Helm and Ship Balloon on it. Use a Ship Dock to isolate the build from your island. Open the Helm, let it scan, and select Captain (Launch). One Balloon supports up to 32 total ship blocks.

Use normal movement controls. Look up or down while moving forward to change altitude. Jump opens the Helm menu while piloting. Use Dock Ship or dismount in a clear area to restore real blocks. Single chests and barrels retain their cargo. Separate the ship from AVA machines before scanning.

Helm I uses a Compass, Bronze, Redstone and Oak Planks. The Helm II recipe uses Helm I with Reinforced Frames, Steel Ingots and Circuit Boards. The Helm III recipe uses Helm II with Reinforced Frames, Advanced Circuits and Conductive Cores. Use the pictured patterns on the next page.

***GO TO*** *Helm and Balloon crafting grids* | *Contents*

# Reactor and ship crafting layouts

Use these recipes in a normal Crafting Table. For the reactor, make the casing and the controller before gathering the cooling blocks and fuel.

**Reactor Controller**



**Reactor Casing**



**Wireless Power Relay**



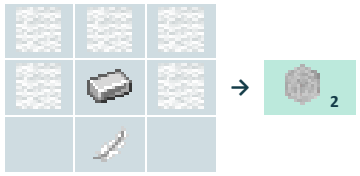
**Uranium Chunk**



**Nuclear Fuel Rod**



**Ship Balloon**



**Ship Helm I**



**Ship Helm II**



**Ship Helm III**



***BUILDS*** *Nuclear power* | *Flying ships*